Generalized Machine Learning for Fast Calibration of Agent-Based Epidemic Models

Abstract

Background. Agent-based models (ABMs) describe how an epidemic spreads through the contacts between individuals, and they are increasingly used to support public health decisions. Before an ABM can be used, its parameters have to be calibrated to the observed data. Standard calibration methods, such as approximate Bayesian computation (ABC) and numerical optimization, run the model thousands of times for every data set, which makes them slow and requires many choices from the user.

Methods. We present DeepIMC (Deep Inverse Mapping Calibration), a neural network that learns the inverse mapping from an epidemic curve to the quantities that produced it. DeepIMC is trained once, on 1,000,000 epidemics simulated with agent-based SEIR and SIR models, and then calibrates a new curve in a single pass of the network. It reads a window of daily new infections of any length between 30 and 365 days, taken from any part of an epidemic, and returns the basic reproduction number R_0 with a 95% range, the transmission probability and contact rate, and the state of the epidemic on the first day of the window. We compared DeepIMC with four standard calibration methods used with their default settings (ABC-MCMC, ABC-SMC, differential evolution and Nelder–Mead) on 10,000 simulated test epidemics, measuring both the accuracy of the estimated parameters and how well each method's estimates reproduce the epidemic curves, and on five real datasets chosen because their counts follow new infections closely. All computing times were measured on CPUs.

Results. On the 10,000 simulated test windows, DeepIMC estimated R_0 with a mean absolute error of 0.17 (SEIR) and 0.14 (SIR), about three times smaller than the best comparison method (0.55 and 0.36), and it was more accurate than that method on 72–82% of the windows. Its 95% ranges contained the true R_0 in 96% of the windows while being three to five times narrower than those of the ABC methods. No method could separate the transmission probability from the contact rate, because only their product shapes the curve. Curves regenerated from DeepIMC's estimates matched the observed new infections as closely as the best comparison method, and recovered the

unobserved number of infectious people with half its error and 98–99% coverage. On real data, DeepIMC fitted the influenza and Ebola series almost as closely as the best comparison method, and was weaker on outbreaks brought under control. On one CPU core, DeepIMC calibrated a window in 0.1 to 0.14 seconds, against 4 to 536 seconds for the comparison methods.

Conclusions. DeepIMC calibrates agent-based epidemic models accurately, reproduces the epidemic they describe, and does so in a fraction of a second, with no settings to choose. Its advantage on real data depends on how closely the data resemble the simulations it was trained on. The method, the trained models and the comparison methods are available in the open-source R package `epiworldRcalibrate`.

Keywords: agent-based models; epidemic modeling; model calibration; machine learning; neural networks; approximate Bayesian computation; simulation-based inference

1 Introduction

Agent-based models (ABMs) simulate an epidemic one person at a time: each agent has a state (susceptible, exposed, infectious or recovered), meets other agents, and may transmit or acquire an infection. Because they represent individuals and their contacts explicitly, ABMs can describe heterogeneous populations, interventions that target particular groups, and the randomness of small outbreaks, which simpler compartmental models cannot. For these reasons they are widely used to explore the course of epidemics and the effect of public health measures.[29]

An ABM is only useful once it is *calibrated*, that is, once its parameters have been chosen so that the model reproduces the observed data. For an epidemic model, the most important of these parameters describe transmission: how many contacts an infectious person has per day, and how likely each contact is to transmit the infection. Calibrating an ABM is computationally demanding. The model has no likelihood that can be written down, so calibration methods must run the model many times and compare its output with the data. Approximate Bayesian computation (ABC) and numerical optimization methods typically need thousands of model runs for each data set, and the whole procedure has to be repeated whenever new data arrive, for every location and every time period of interest. During an outbreak, when estimates are needed quickly and repeatedly, this cost limits how ABMs can be used.

A second difficulty is that standard calibration methods require many decisions from the user: the prior ranges of the parameters, how to compare simulated and observed curves (the whole curve or a set of summary statistics), tolerances, the number of iterations or particles,

and the starting values. These choices affect the results, and making them well requires expertise in both the model and the calibration method.

Machine learning offers a different route. Rather than searching the parameter space for every new data set, a neural network can be trained once, on simulations from the model, to map the observed data directly to the parameters that generated them. This idea, often called *amortized* or *simulation-based* inference,[7,28] moves the computational cost to a one-time training phase; afterwards, calibrating a new data set takes a single pass of the network. Most applications of machine learning to ABM calibration have instead built surrogate models, which approximate the mapping from parameters to model outputs and still require a search over parameters for every data set.[2,17]

In this paper we develop and evaluate DeepIMC (Deep Inverse Mapping Calibration), a neural network that calibrates agent-based SEIR and SIR epidemic models from the **epiworldR** package.[23] DeepIMC is designed for the data available in practice. Analysts rarely observe a complete epidemic from its first case: they work with the last weeks or months of data, which may start in the middle of a wave. DeepIMC therefore reads a window of daily new infections of any length between 30 and 365 days, from any part of an epidemic, and estimates, besides the parameters, the state of the epidemic on the first day of the window: how many people are exposed, infectious and recovered. This state is what is needed to simulate the epidemic forward.

Our contributions are the following.

1. We describe DeepIMC, which estimates the basic reproduction number R_0, the transmission probability, the contact rate and the current state of an agent-based epidemic from a window of daily new infections, together with a 95% range for R_0.

2. We compare DeepIMC with four standard calibration methods—ABC-MCMC, ABC-SMC, differential evolution and Nelder–Mead—used as a practitioner would use them, on 10,000 simulated test epidemics. We evaluate not only the accuracy of the estimated parameters but also how well each method's estimates reproduce the epidemic: the observed curve, the number of people currently infectious (which no method observes), and the following two weeks.

3. We evaluate all methods on five real datasets, which we selected by checking what each dataset counts and whether those counts follow new infections closely enough to be modelled.

4. We compare the computing time of all methods on CPUs, separating the one-time cost of training DeepIMC from the cost of calibrating a new data set.

5. We provide the method, the trained models, the comparison methods and the evaluation tools in the open-source R package `epiworldRcalibrate`.

The rest of the paper is organized as follows. Section 2 reviews calibration methods for agent-based models. Section 3 describes the models, DeepIMC, the comparison methods and the evaluation. Section 4 presents the results on simulated and real data and the computing times, and Section 5 discusses what they mean for the use of agent-based models in practice.

2 Calibrating agent-based models

Agent-based models. An agent-based model consists of agents, the rules by which they interact (the topology of their contacts), and their environment.[29] In an epidemic ABM, each agent is in one disease state at a time, and the transitions between states happen at random according to the model's parameters and the agent's contacts. The behaviour of the whole population—the epidemic curve—emerges from these individual events. Because the model is stochastic, running it twice with the same parameters gives two different epidemics.

The calibration problem. Calibration seeks the parameter values for which the model's output resembles the observed data. For an ABM the probability of the observed data given the parameters (the likelihood) cannot be computed, so calibration methods compare simulated and observed data directly. Strategies include approximate Bayesian computation,[31,12] simulated minimum distance and other optimization methods,[26] and surrogate models.[16,14,36]

Approximate Bayesian computation. ABC approximates the posterior distribution of the parameters by keeping parameter values whose simulated data are close to the observed data. The data are usually reduced to summary statistics, and closeness is measured by a distance between the observed and simulated statistics.[3] In its simplest form, ABC draws parameters from the prior and accepts those within a tolerance; this is very inefficient when the prior is wide. ABC-MCMC[22] explores the parameter space with a Markov chain, proposing small moves from the current value, and ABC-SMC[31,4] moves a population of particles through a sequence of decreasing tolerances. Both return samples from an approximate posterior, and both need thousands of model runs per data set.

Optimization. Optimization methods search for the single parameter value whose simulated output is closest to the data. Differential evolution[30] evolves a population of candidate solutions by combining them, and is robust to local minima; the Nelder–Mead simplex method[25] moves a small set of points downhill and needs far fewer model runs, but can stop

early when the objective is flat or noisy, as it is for a stochastic model. Optimization gives a point estimate, without a measure of uncertainty.

Surrogates and amortized inference. A surrogate (or emulator) is a fast statistical model of the mapping from parameters to model output, fitted to a set of model runs; it speeds up each evaluation, but calibration still searches the parameter space for every data set. *Amortized* inference instead learns the inverse mapping, from data to parameters, once, from simulations.[7,28] After training, calibrating a new data set costs one evaluation of the learned mapping. DeepIMC follows this approach, with a recurrent neural network that reads the epidemic curve day by day.

3 Methods

3.1 The agent-based models

We consider the two connected (homogeneous mixing) models of **epiworldR**: `ModelSEIRCONN`, in which agents are susceptible, exposed, infectious or recovered, and `ModelSIRCONN`, without the exposed state. Each day, every infectious agent meets on average c_{rate} randomly chosen agents (the *contact rate*), and each meeting with a susceptible agent transmits the infection with probability p_{tran} (the *transmission probability*). In the SEIR model, an infected agent first becomes exposed and moves to the infectious state with daily probability 1/incub, so that incub is the mean latent period in days. Infectious agents recover with daily probability p_{recov}, so that $1/p_{recov}$ is the mean infectious period. The basic reproduction number, the mean number of people infected by one infectious person in a fully susceptible population, is

$$R_0 = \frac{p_{tran}\, c_{rate}}{p_{recov}}. \tag{1}$$

What a curve can reveal. In these models the chance that a susceptible agent is infected on a given day depends on p_{tran} and c_{rate} only through their product, $p_{tran} \times c_{rate}$, the daily transmission rate. Two pairs with the same product produce epidemics with the same distribution: for example, $p_{tran} = 0.071$ with $c_{rate} = 5$, and $p_{tran} = 0.024$ with $c_{rate} = 15$, give the same epidemic. A curve of daily new infections can therefore reveal R_0 (given the known recovery rate), but not how it splits into p_{tran} and c_{rate}. We report the estimates of all three parameters, and compare the errors in p_{tran} and c_{rate} with those of a guess that knows R_0 exactly but not the split.

3.2 The calibration problem

The data. The observed data are a *window*: the daily counts of new infectious people (transitions from exposed to infectious in the SEIR model, from susceptible to infectious in the SIR model) over L consecutive days, with $30 \leq L \leq 365$. The window may come from the beginning, the middle or the end of an epidemic.

The known values. As is common in applied epidemic modelling, we assume that some quantities are known or can be taken from other studies: the population size N, the recovery rate p_{recov} and, for the SEIR model, the latent period incub.

The unknowns. DeepIMC estimates R_0, p_{tran} and c_{rate}, and the state of the epidemic on the first day of the window: the shares of the population exposed (SEIR), infectious and recovered. The state matters as much as the parameters: a window from the middle of an epidemic starts with many people already infected or immune, and the same parameters lead to very different futures depending on that state.

3.3 Training data

For each model we simulated 1,000,000 epidemics of 365 days. The parameters of each epidemic were drawn independently from the distributions in Table 1, chosen to cover a wide range of respiratory infections: R_0 between 0.8 and 5, recovery rates between 0.071 and 0.25 (an infectious period of 4 to 14 days), latent periods of 1 to 21 days, and populations of 5,000 to 10,000. Each epidemic started with a share of the population exposed (SEIR) or infectious (SIR) drawn on the logarithmic scale between 0.1% and 40%, so that both small outbreaks and epidemics already well under way are represented. For every day we stored the number of new infectious people and the shares of the population in each state. Epidemics with fewer than 20 cases in total were discarded (0.1% for SEIR and 0.2% for SIR). These data are noise-free: every new infection is counted.

Windows. The network is trained on windows cut from these epidemics, not on whole epidemics. Every training batch cuts one window from each epidemic in the batch. All windows of a batch share one length, drawn uniformly between 30 and 365 days, and each window starts at a random day within the *informative* part of its epidemic: the days before the first run of seven days without new infections. Later days carry no information and are never observed in practice. A new set of windows is drawn every epoch, so the network sees each epidemic through many different windows.

Table 1: Distributions used to simulate the training, calibration and test epidemics.

Quantity	SEIR	SIR
Population size N (known)	uniform, 5,000 to 10,000	same
Recovery rate p_{recov} (known)	uniform, 0.071 to 0.25	same
Latent period incub (days, known)	uniform, 1 to 21	–
Basic reproduction number R_0	uniform, 0.8 to 5	same
Contact rate c_{rate}	uniform, 5 to 15	uniform, 1 to 5
Transmission probability p_{tran}	$R_0\, p_{recov}/c_{rate}$	same (at most 1)
Initial share exposed (SEIR) or infectious (SIR)	log-uniform, 0.001 to 0.4	same
Days simulated	365	365

3.4 The network

Why a recurrent network. A window is a sequence of daily counts whose length varies from one data set to the next. A long short-term memory (LSTM) network[13] reads such a sequence one day at a time and keeps a summary of what it has seen, so the same network can read windows of any length. We use a *bidirectional* LSTM, which reads the window both forwards and backwards, so that the summary of every day uses the whole window; this helps the network recognise whether the window covers the growth, the peak or the decline of an epidemic.

Inputs. The daily counts are transformed as $\log(1+x_t)/\log(1+10{,}000)$, which puts small and large epidemics on a comparable scale. The known values (N, p_{recov} and, for SEIR, incub) are scaled to the interval from 0 to 1.

Architecture. Figure 1 shows the network. The final states of the forward and backward LSTM are joined with the known values and passed through a dense layer of 64 units and an output layer with one unit per target. The SEIR network has three LSTM layers of 128 units in each direction; the SIR network, two layers of 64 units. These settings were chosen by a hyperparameter search (below).

Outputs. The network predicts p_{tran}, R_0 and the logarithm of each state share on the first day of the window. The contact rate is then computed from (1) as $c_{rate} = R_0\, p_{recov}/p_{tran}$, so the three parameters always agree with each other.

Epidemiological information. A network that predicts p_{tran}, c_{rate} and R_0 as three separate outputs can return values that contradict (1); a common remedy is to add a penalty such as $\left(R_0\, p_{recov} - p_{tran}\, c_{rate}\right)^2$ to the loss, which pushes the outputs towards agreement

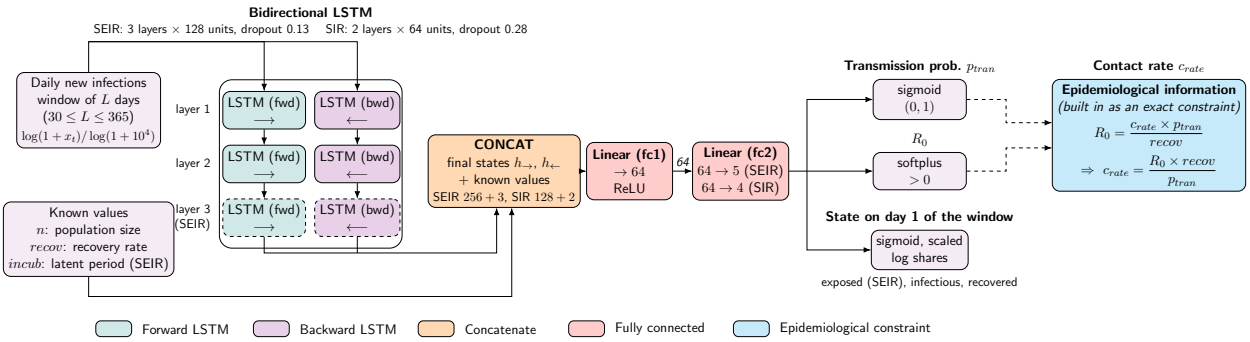


Figure 1: The DeepIMC network. A bidirectional LSTM reads the window of daily new infections forwards and backwards; its final states are joined with the known values and passed through two dense layers, which give the transmission probability, R_0 and the state of the epidemic on the first day of the window. The epidemiological relation $R_0 = c_{rate}\, p_{tran}/recov$ is built in as an exact constraint: the contact rate is computed from R_0 and p_{tran}, so the three parameters always agree. The shipped models average five such networks, trained with different random seeds, and attach a 95% range to R_0 from 500 similar calibration windows (Section 3.5).

without guaranteeing it. DeepIMC builds this relation into the network instead: because c_{rate} is computed from R_0 and p_{tran}, the relation holds exactly for every window, which is equivalent to a penalty with infinite weight. The penalty term is then always zero and is not needed in the loss.

Training. The network was trained to minimize the mean squared error of the scaled targets, with the Adam optimizer,[15] batches of 32 windows, and a learning rate that starts at 7.2×10^{-4} (SEIR) or 7.9×10^{-4} (SIR) and decreases along a cosine curve to 1% of that value over 50 epochs.[21] Twenty percent of the epidemics were held out for validation, and the weights of the epoch with the lowest validation error were kept. We trained five networks with different random seeds and average their outputs.

Hyperparameter search. The learning rate, batch size, number of layers, number of units and dropout were chosen with Optuna,[1] in 40 trials of 20 epochs on 100,000 of the training epidemics (Appendix B).

3.5 Uncertainty in R_0

A single pass of the network gives point estimates. To attach a range to R_0, we simulated 100,000 further epidemics, cut one random window from each, and recorded DeepIMC's error on each of these *calibration windows*. For a new window, we find the 500 calibration windows

that are most similar to it—with similar estimates of R_0 and of the starting state, similar known values and a similar length—and use the spread of their errors. The 2.5% and 97.5% quantiles of those errors, added to the estimate, give the 95% range of R_0. The same 500 errors, in R_0 and in the state shares, give 500 plausible parameter sets, which are used to simulate the epidemic forward.

3.6 From estimates to epidemic curves

Model simulation. To see what an estimate implies for the epidemic, we run the agent-based model 500 times from the estimated state on the first day of the window, each run with parameters drawn from the method's uncertainty: DeepIMC's 500 parameter sets, the ABC methods' posterior samples, or the single estimate of differential evolution and Nelder–Mead. For every day, the 2.5%, 50% and 97.5% quantiles of the runs give the regenerated curve (the median) and its 95% band, both in the window and for the following days.

The renewal equation. A much faster alternative is the renewal equation,[10,34] which gives the expected number of new infectious people on day t as

$$C_t = R_0 \, \frac{S_{t-1}}{N} \sum_{k \geq 1} g_k \, C_{t-k}, \tag{2}$$

where S_t is the number susceptible and g_k the generation interval of the model, the probability that a transmission happens k days after the infector was infected. Because the latent and infectious periods of our models are geometric, the sum can be carried forward day by day (Appendix G). The renewal equation gives the model's average curve; we add Poisson noise around it to obtain a band. We start it either from DeepIMC's estimated state (renewal A) or from a state that the renewal equation itself fits to the window by least squares, with R_0 fixed at 30 values from DeepIMC's range (renewal B). The main results use the agent-based model; the renewal equation is evaluated as a faster alternative in the supplement (Appendix F).

3.7 Comparison methods

We compared DeepIMC with four standard calibration methods, set up as a practitioner with basic knowledge of them would use them: established implementations, their default settings, and only the changes needed for them to work with these models (Table 2, Appendix C). Using default settings is deliberate: one of DeepIMC's aims is to spare the user these choices, so the comparison shows what a user obtains without tuning.

The same problem for every method. All methods receive the same window and known values and search the same unknowns within the same ranges as DeepIMC: R_0 between 0.8 and 5, and the state on the first day of the window (the shares exposed, infectious and recovered, searched on the logarithmic scale between 10^{-4} and 0.4, or 0.95 for the recovered share). Because the curve reveals R_0 but not the split into p_{tran} and c_{rate} (Section 3.1), the comparison methods fix c_{rate} at the middle of its range and obtain p_{tran} from (1).

Comparing a simulated curve with the data. Following common practice in approximate Bayesian computation,[3] ABC-MCMC and ABC-SMC compare *summary statistics* of the observed and simulated curves: the mean, the peak and the minimum of the daily counts, the day of the peak, the means of the first and the last week, the days by which 25%, 50% and 75% of the window's cases had occurred, and the growth rates over the first and the last two weeks (Appendix C). Each statistic is divided by its spread over simulations from the prior, so that statistics measured in different units count equally.[27] Differential evolution and Nelder–Mead minimize the sum of squared differences between the observed counts and the simulated daily new infections, using one model run with a fixed random seed per candidate so that the same candidate always receives the same fit. Appendix D compares the ABC methods on summary statistics with the same methods on the whole curve.

Necessary adjustments. With its default limits of 8,000 model runs and 5,000 attempts per generation, ABC-SMC stopped before finishing on most simulated SEIR windows; we raised the two limits, as its error message suggests. Differential evolution always runs its 200 generations, because its default early-stopping rule (no improvement for 200 generations) cannot be met within 200 generations.

3.8 Evaluation on simulated data

Simulated data are the only setting in which the true parameters and the true epidemic are known, so most of the evaluation uses them. All test epidemics were simulated with the distributions of Table 1 but with new random seeds, and none was used in training.

DeepIMC alone, by window length. To show how DeepIMC's accuracy depends on how much data it sees, we simulated 1,000 test epidemics per model and evaluated DeepIMC on windows of 30, 61, 90, 120, 180, 270 and 365 days starting at the beginning (day 1), the middle and the end (the last informative day) of each epidemic that lasts that long.

Table 2: The comparison methods and their settings. All are run with their default settings, except the two limits of ABC-SMC (see text).

Method	Settings	Model runs per window	Stopping rule
DeepIMC	average of 5 networks; R_0 range from 500 similar calibration windows	0	one pass of the networks
ABC-MCMC[22]	`epiworldRcalibrate`: summary statistics; 3,000 random-walk steps (1,500 burn-in), proposal s.d. 8% of each range, Gaussian kernel with the tolerance set from 200 prior simulations	about 3,200	after 3,000 steps
ABC-SMC[31]	`epiworldRcalibrate`: summary statistics; 200 particles, 6 generations, each tolerance the median distance of the previous generation; at most 30,000 model runs and 20,000 attempts per generation	up to 30,000	after 6 generations
DE[30]	`DEoptim`:[24] population of 10 times the number of unknowns, 200 generations, $F = 0.8$, $CR = 0.5$; sum of squared differences	6,030–8,040	after 200 generations
Nelder–Mead[25]	`stats::optim`: started at the middle of the ranges, values outside the ranges rejected; sum of squared differences	about 100	relative change of the fit below 1.5×10^{-8}, or 500 runs

All methods: 10,000 test windows. For the comparison with the other methods, we simulated 10,000 further test epidemics (5,000 per model) and took one window from each: a random length between 30 and 180 days at a random start among the epidemic's informative days (epidemics with fewer than 30 informative days were skipped). We call a window *early*, *middle* or *late* when it starts in the first, second or last third of the possible starts. Every method calibrated every window.

Parameter recovery. For each parameter we report the mean absolute error (MAE) over the windows, $\frac{1}{n}\sum_i |\hat{\theta}_i - \theta_i|$, where θ_i is the true value and $\hat{\theta}_i$ the estimate, for R_0, p_{tran} and c_{rate}. The starting-state error is the mean absolute error of the state shares, in percentage points. For DeepIMC and the ABC methods, which give a 95% range for R_0, we also report how often the true value lies inside it; a well-calibrated range should contain it about 95% of the time.

Recovery of the epidemic curves. A method can estimate parameters well and still reproduce the epidemic poorly, or the reverse. We therefore used each method's estimates to regenerate the epidemic (500 model runs, Section 3.6) for the window and the 14 days after it, and compared the regenerated curves with the true curves of the test epidemic in three ways:

- the daily new infections in the window, the data each method saw;
- the number of people currently infectious in the window, which no method observes;
- the daily new infections over the 14 days after the window, a short-term prediction.

For each, we report the coverage of the 95% band and the MAE of the regenerated median curve (Section 3.10).

3.9 Evaluation on real data

Choosing appropriate datasets. Real data have no true parameter values, so on real data the methods are compared by how well the curves regenerated from their estimates match the observed counts. For this comparison to be meaningful, the data should measure what the models describe: the daily number of new infections. We therefore examined what each candidate dataset counts. Case counts by date of test report follow testing and reporting as much as infection: in the Utah COVID-19 cases, for example, Sundays are 30–40% below the weekly average, and after home testing became common most infections were not reported.[33] Counts by date of symptom onset, and hospital admissions by date of

admission, follow new infections more closely: they are not shaped by the reporting week, and a stable fraction of infections leads to them after a fairly constant delay. We selected five such datasets (Table 3), covering four diseases, and kept the Utah COVID-19 cases by test report date as a contrast.

Table 3: The real datasets: what each counts, its source, and the known values used for each disease (SEIR latent period incub in days and recovery rate; SIR recovery rate).

Dataset	What is counted (by date of)	Period and source	Windows per model	Known values (SEIR; SIR)
Utah COVID-19	confirmed COVID-19 hospital admissions (admission)	three waves, Sep 2020–Mar 2022; Utah DHHS[33]	27	incub 5.2, p_{recov} 0.25; p_{recov} 0.133[19]
Utah influenza	confirmed influenza hospital admissions (admission)	2022–23 and 2023–24 seasons; US HHS[32]	18	incub 1.4, p_{recov} 0.25; p_{recov} 0.25[18,5]
Baltimore influenza 1918	influenza cases from a house-to-house survey (symptom onset)	92 days, autumn 1918[11,5]	9	as Utah influenza
Hong Kong SARS 2003	SARS cases (symptom onset)	107 days, 2003[6]	9	incub 6.4, p_{recov} 0.25; p_{recov} 0.119[8,20]
Sierra Leone Ebola 2014–15	confirmed and suspected cases (symptom onset)	May 2014–Sep 2015[9]	9	incub 11.4, p_{recov} 0.25; p_{recov} 0.071[35]
Contrast: Utah COVID-19 cases	confirmed cases (test report)	as Utah COVID-19	27	as Utah COVID-19

Known values. The known values for each disease come from published estimates of the mean incubation period and the mean serial interval (the time between the symptom onsets of an infector and the person they infect). In the SEIR model, the latent period is set to the incubation period, and the infectious period to the serial interval minus the incubation period, but at least 4 days, the shortest in the training data. In the SIR model, which has no latent period, the infectious period is set to the serial interval, within the trained range of 4 to 14 days.

Population size. The models describe an epidemic in which every new infection is counted, in a population of 5,000 to 10,000. Real counts cover only a fraction of infections in a much larger population, and the reported fraction and the population size cannot be separated

from a curve alone. We therefore read every dataset as the complete incidence of an effective population of $N = 10{,}000$. The Utah COVID-19 cases are far more than 10,000 people could produce and were scaled to cases per 10,000 residents.

Windows. In each wave or epidemic we took windows of 30, 60 and 90 days at the beginning, the middle (centred on the peak of the 7-day average) and the end, and scored the bands on the window and on the next 14 days where the data continue.

3.10 Scores

A 95% band $[l_t, u_t]$ and its median curve m_t are compared with the values y_t (true or observed) over T days by two scores.

- *Coverage*, the percentage of days on which the value lies inside the band, $\frac{100}{T}\sum_t \mathbf{1}(l_t \le y_t \le u_t)$. About 95% is ideal for a 95% band; much lower means the band misses the data, and 100% can mean the band is too wide.
- *Mean absolute error* (MAE) of the median curve, $\frac{1}{T}\sum_t |m_t - y_t|$, in daily counts (or, for the number currently infectious, in people). Lower is better.

The two scores answer different questions. Coverage asks whether the band is honest about its uncertainty; the MAE asks how close the method's best guess is to the truth. A method can reach high coverage with very wide bands while its median curve is far from the data, so we always report both. For the same reason we report, besides the coverage of the R_0 ranges, their average width.

3.11 Computing time

All timings were made on CPUs only, without a GPU, so that the methods are compared on the same hardware. Every method ran on one CPU core of the same computer (an AMD EPYC 7601 processor), on 20 windows of 30 to 180 days (10 per model) from the simulated test set. For DeepIMC we measured its first call in a new R session, through the package's functions (which start Python and load the networks), one window at a time, and many windows in one pass. We also measured the cost of turning a fit into a 95% band. Training DeepIMC is a separate, one-time cost: we report the CPU time to simulate the training epidemics and the time to train one network, on CPUs (estimated from one training epoch) and on the GPUs we actually used.

4 Results

4.1 How DeepIMC's accuracy depends on the window

The first question is how much data DeepIMC needs. Table 4 shows its errors on windows of increasing length that start at the beginning of the test epidemics.

The error in R_0 falls quickly as the window grows. With 30 days of data the mean absolute error is 0.28 for the SEIR model and 0.14 for the SIR model; with 180 days it is 0.06 and 0.05. The SEIR model needs more data than the SIR model, because the latent period delays the effect of transmission on the curve. Windows that start in the middle or at the end of an epidemic are harder than those that start at the beginning, most of all when they are short (column "any position"): in these windows part of the population is already immune, and the network must also work out how many people have been infected before the window begins.

Figure 2 shows the same results by position. The error is largest for short windows at the end of an epidemic (0.76 for 30-day SEIR windows), where little of the epidemic is left to see; it falls below 0.2 once the window is 120 days long, and for windows of 270 days or more the position no longer matters. The coverage of the true curves stays above 94% for every length and position, except for the true number of infectious people in 30-day SEIR windows at the beginning of an epidemic (90%).

The 95% range of R_0 contained the true value in 88–99% of the SEIR windows and 92–100% of the SIR windows, depending on the length; over all windows, 94% for both models. The range is slightly too narrow for the shortest SEIR windows and well calibrated otherwise.

The transmission probability and the contact rate. The errors in p_{tran} and c_{rate} behave differently from the error in R_0: they hardly change with the window length, and they are almost exactly those of the guess that knows R_0 and picks a typical split (Table 4). For 61-day SEIR windows, for example, DeepIMC's p_{tran} error is 0.0145 against 0.0144 for the guess, and its c_{rate} error 2.59 against 2.56. This is what Section 3.1 predicts: the curve contains information about R_0, the product of the two, but none about how it splits. No calibration method can do better from case curves alone.

The epidemic behind the window. When the model is run forward from DeepIMC's estimates, the true daily new infections fall inside the 95% band on 96–99% of the days at every window length, and the true number of infectious people—which the network never sees—on 94–99% of the days. DeepIMC therefore recovers not only R_0 but also the hidden state of the epidemic.

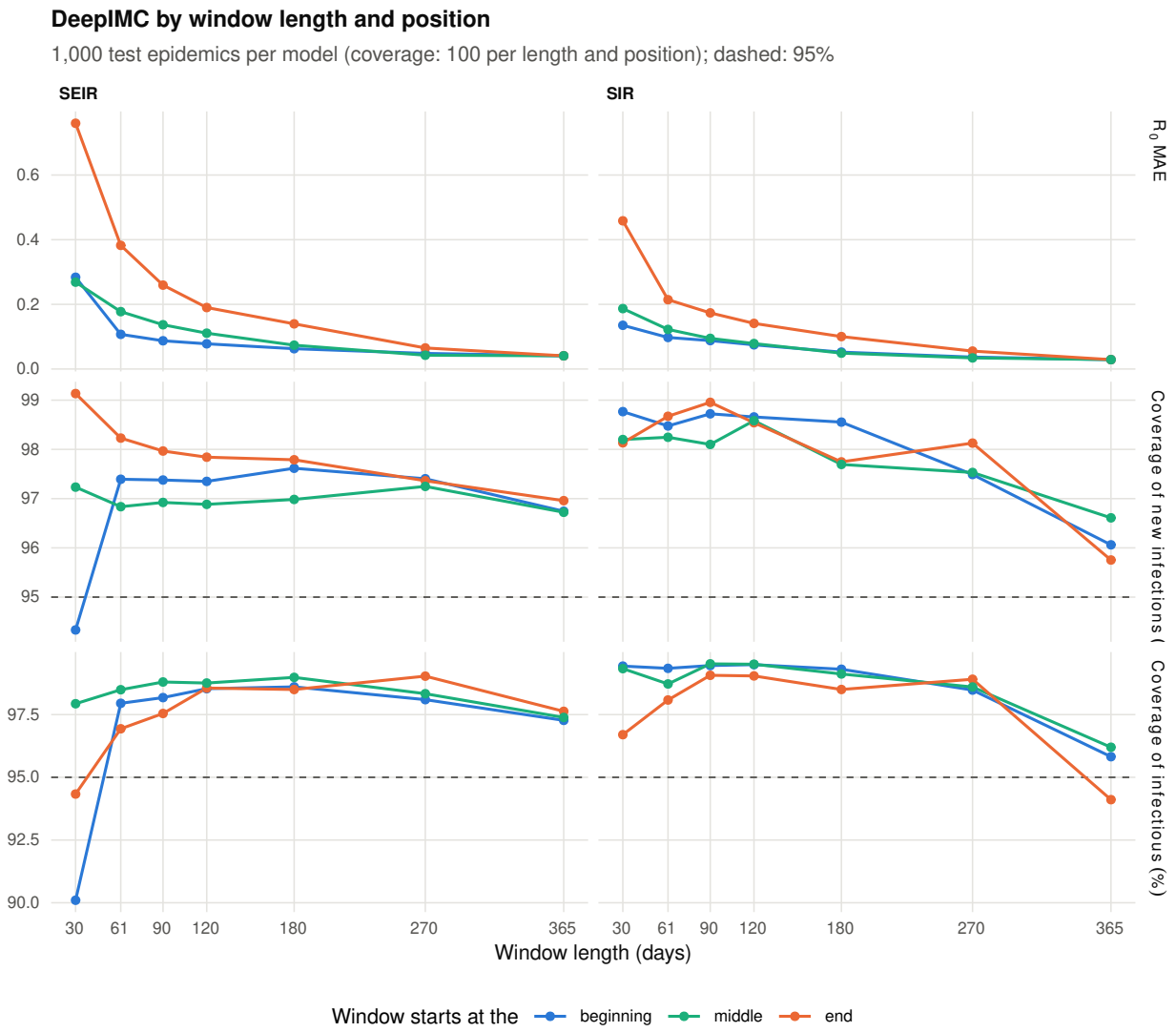


Figure 2: DeepIMC by window length and position: R_0 mean absolute error, and coverage of the true daily new infections and of the true number currently infectious by the 95% band, for windows that start at the beginning, the middle and the end of the test epidemics.

Table 4: DeepIMC on simulated windows that start at the beginning of the test epidemics, by window length. MAE of R_0 for these windows and for windows at any position (beginning, middle and end); percentage of windows whose 95% range contains the true R_0; MAE of p_{tran} and c_{rate}, with in brackets the error of a guess that knows R_0 exactly but not how it splits into p_{tran} and c_{rate}; coverage of the true daily new infections and of the true number currently infectious by the 95% band of 100 simulations from the estimates, over windows at any position (100 test epidemics per length and position).

	Window		R_0 MAE		True R_0	p_{tran} MAE	c_{rate} MAE	Coverage (%)	
Model	(days)	Windows	beginning	any position	in range (%)	(guess)	(guess)	new inf.	infectious
SEIR	30	990	0.284	0.438	87.7	0.0157 (0.0147)	2.60 (2.56)	96.9	94.1
	61	960	0.107	0.222	94.7	0.0145 (0.0144)	2.59 (2.56)	97.5	97.8
	90	883	0.087	0.161	95.7	0.0139 (0.0138)	2.59 (2.55)	97.4	98.2
	120	746	0.078	0.126	96.5	0.0131 (0.0130)	2.55 (2.52)	97.4	98.6
	180	475	0.063	0.092	95.2	0.0109 (0.0108)	2.47 (2.47)	97.5	98.7
	270	178	0.048	0.052	97.1	0.0075 (0.0074)	2.38 (2.37)	97.3	98.5
	365	80	0.041	0.041	98.9	0.0073 (0.0071)	2.65 (2.57)	96.8	97.4
SIR	30	982	0.135	0.260	93.6	0.0643 (0.0685)	1.01 (0.97)	98.4	98.5
	61	572	0.098	0.145	94.6	0.0486 (0.0482)	1.07 (1.00)	98.5	98.7
	90	293	0.088	0.119	95.0	0.0364 (0.0356)	1.09 (1.02)	98.6	99.3
	120	159	0.075	0.098	93.9	0.0329 (0.0319)	1.14 (1.04)	98.6	99.3
	180	53	0.052	0.067	91.8	0.0280 (0.0280)	1.14 (1.08)	98.0	99.0
	270	18	0.037	0.042	95.2	0.0239 (0.0242)	0.99 (1.02)	97.7	98.7
	365	8	0.029	0.029	100.0	0.0153 (0.0150)	0.91 (0.83)	96.1	95.4

4.2 Parameter recovery compared with standard methods

We now turn to the comparison with the four standard methods on the 10,000 simulated test windows. The first question is how accurately each method recovers the parameters and the starting state.

R_0. DeepIMC estimated R_0 with a mean absolute error of 0.17 in the SEIR windows and 0.14 in the SIR windows (Table 5). The best comparison method was ABC-SMC for SEIR (0.55) and differential evolution for SIR (0.36), so DeepIMC's error was about three times smaller for SEIR and 2.6 times smaller for SIR. ABC-MCMC (0.81 for both models) and Nelder–Mead (0.95 and 0.65) were the least accurate. On the same windows, every comparison method had a larger error than DeepIMC, for every position of the window, and none of the 95% bootstrap intervals of the paired differences includes zero (Table 6). DeepIMC was more accurate than ABC-SMC on 82% (SEIR) and 80% (SIR) of the windows, and more accurate than differential evolution on 75% and 72%.

The position of the window. Where the window comes from matters a great deal for the comparison methods. On early windows, which contain the growth of the epidemic, differential evolution came closest to DeepIMC (0.24 against 0.12 for SEIR, 0.17 against 0.11

for SIR). On middle and late windows, where a method must also work out how many people were already infected or immune, the errors of the comparison methods were two to four times larger than on early windows, while DeepIMC's errors grew much less (to 0.16 and 0.25 for SEIR, 0.13 and 0.18 for SIR). Figure 3 breaks these results down by window length as well: DeepIMC had the lowest R_0 error in every group of length and position, and its advantage was largest for short windows in the middle and at the end of an epidemic, where the comparison methods' errors were largest. Longer windows helped every method except ABC-MCMC, whose error did not fall with the length of the window, and in SIR windows grew from 0.6–0.8 to 1.5–1.7. Figure 4 shows how the comparison methods fail: ABC-MCMC's estimates are pulled towards the middle of the range, ABC-SMC underestimates many large values of R_0, and differential evolution and Nelder–Mead return many estimates at the lower limit of 0.8 whatever the true value.

The range of R_0. DeepIMC's 95% range contained the true R_0 in 96% of the windows for both models, close to the nominal 95%, with an average width of 0.90 (SEIR) and 0.70 (SIR). The ranges of ABC-MCMC and ABC-SMC contained it about as often (96–98%), but they were three to five times wider (2.6 to 3.5, for a prior range of width 4.2): they were rarely wrong because they said little.

The transmission probability and the contact rate. As expected from Section 3.1, no method recovered p_{tran} and c_{rate} separately (Figure 4, rows p_{tran} and c_{rate}). DeepIMC's errors were close to those of the guess that knows R_0 exactly and takes the middle of the range of the contact rate: 0.0147 against 0.0140 for p_{tran} in SEIR windows, and 2.59 against 2.51 for c_{rate}. The comparison methods fix c_{rate} at the middle of its range, so their c_{rate} error is that of the guess by construction; their p_{tran} errors are larger than DeepIMC's because their R_0 is less accurate.

The starting state. The error of the estimated shares exposed, infectious and recovered at the start of the window was 1.4 (SEIR) and 2.2 (SIR) percentage points for DeepIMC, against 4.1 to 9.7 and 4.5 to 10.0 for the comparison methods. ABC-SMC failed on 9 SEIR windows even with its raised limits; they are left out of its averages.

4.3 Recovery of the epidemic curves

The second question is whether each method's estimates reproduce the epidemic itself.

For each window, we ran the agent-based model 500 times from each method's estimates and compared the regenerated curves with the true curves of the test epidemic (Table 7,

Table 5: Parameter recovery on the 10,000 simulated test windows (5,000 per model). MAE of R_0 for all windows and for early, middle and late windows; for the methods that give a 95% range of R_0, the percentage of windows whose range contains the true value and the average width of the range; MAE of p_{tran} and c_{rate}, with the error of a guess that knows R_0 exactly and takes the middle of the range of c_{rate}; starting-state error (percentage points). Bold: lowest error. The comparison methods fix c_{rate} at the middle of its range, so their c_{rate} error is that of the guess.

		R_0 MAE				95% range of R_0		p_{tran}	c_{rate}	State
Model	Method	all	early	middle	late	has true (%)	width	MAE	MAE	error
SEIR	DeepIMC	**0.17**	**0.12**	**0.16**	**0.25**	95.9	0.90	0.0147	2.59	**1.4**
	ABC-MCMC	0.81	0.69	0.82	0.93	95.6	3.55	0.0192	2.51	9.2
	ABC-SMC	0.55	0.33	0.52	0.80	97.5	2.99	0.0170	2.51	4.9
	DE	0.62	0.24	0.64	0.99	–	–	0.0187	2.51	4.1
	Nelder–Mead	0.95	0.47	0.96	1.42	–	–	0.0223	2.51	9.7
	guess knowing R_0							0.0140	2.51	
SIR	DeepIMC	**0.14**	**0.11**	**0.13**	**0.18**	95.8	0.70	0.0685	1.05	**2.2**
	ABC-MCMC	0.81	0.73	0.83	0.87	96.4	3.33	0.0810	1.01	10.0
	ABC-SMC	0.40	0.27	0.34	0.57	98.4	2.57	0.0709	1.01	5.4
	DE	0.36	0.17	0.39	0.54	–	–	0.0714	1.01	4.5
	Nelder–Mead	0.65	0.24	0.68	1.05	–	–	0.0807	1.01	8.2
	guess knowing R_0							0.0716	1.01	

Table 6: Paired difference in the R_0 absolute error, comparison method minus DeepIMC, on the same windows (positive: DeepIMC is more accurate), with 95% bootstrap intervals.

Model	Method	All windows	Early	Middle	Late
SEIR	ABC-MCMC	0.64 (0.62, 0.65)	0.57 (0.54, 0.59)	0.66 (0.63, 0.69)	0.69 (0.65, 0.72)
	ABC-SMC	0.38 (0.36, 0.39)	0.21 (0.20, 0.23)	0.36 (0.34, 0.39)	0.56 (0.52, 0.59)
	DE	0.45 (0.43, 0.47)	0.12 (0.11, 0.14)	0.48 (0.45, 0.52)	0.74 (0.70, 0.79)
	Nelder–Mead	0.77 (0.75, 0.80)	0.36 (0.33, 0.38)	0.80 (0.76, 0.85)	1.18 (1.12, 1.23)
SIR	ABC-MCMC	0.67 (0.65, 0.68)	0.62 (0.60, 0.65)	0.70 (0.67, 0.73)	0.69 (0.66, 0.72)
	ABC-SMC	0.25 (0.24, 0.27)	0.16 (0.15, 0.17)	0.21 (0.19, 0.23)	0.39 (0.36, 0.42)
	DE	0.22 (0.20, 0.23)	0.06 (0.04, 0.07)	0.26 (0.24, 0.28)	0.35 (0.33, 0.38)
	Nelder–Mead	0.51 (0.48, 0.53)	0.13 (0.11, 0.14)	0.55 (0.50, 0.59)	0.87 (0.82, 0.92)

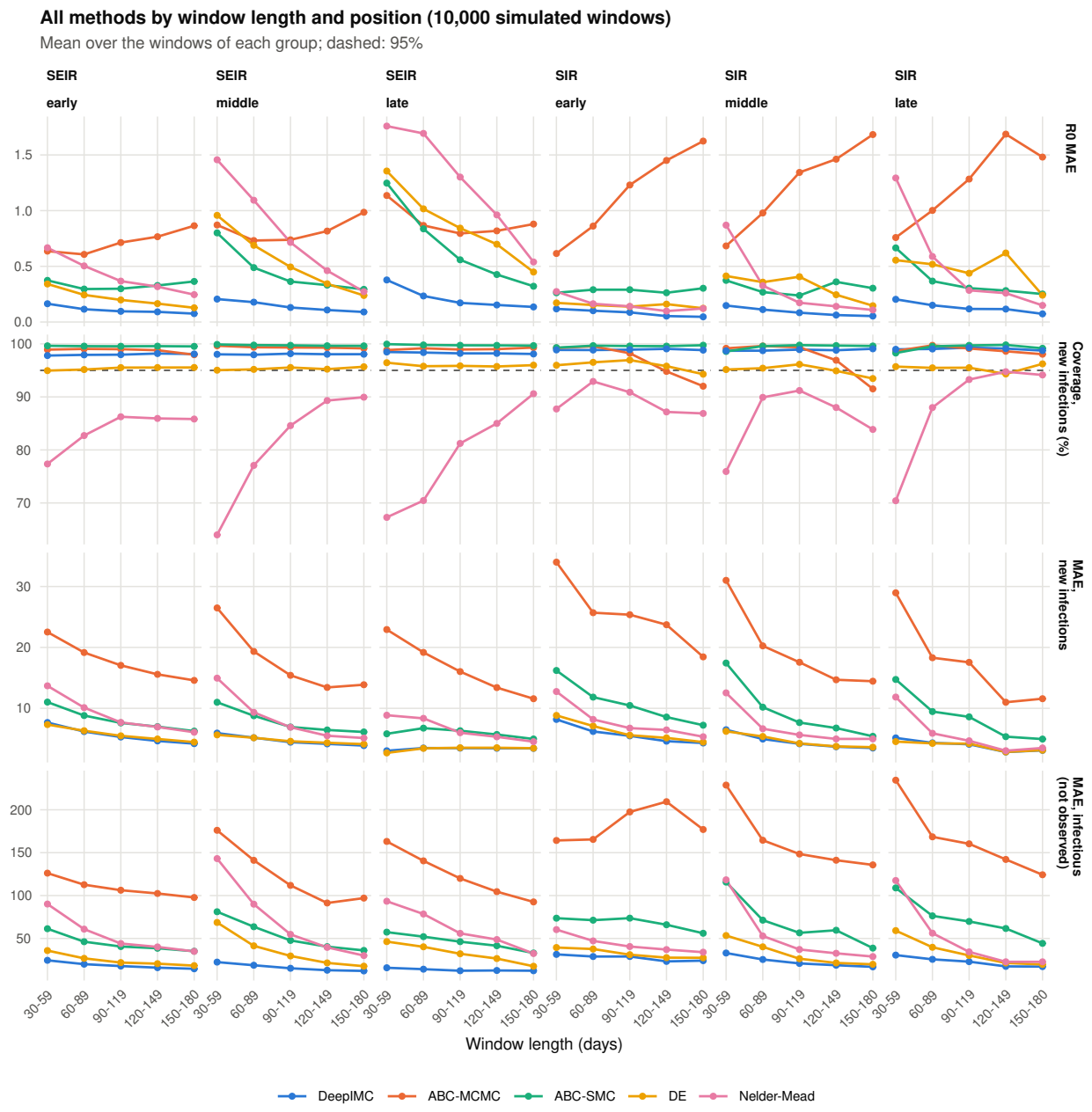


Figure 3: All methods on the 10,000 simulated test windows, by window length (groups of 30 days) and position (early, middle, late): R_0 mean absolute error; coverage of the true daily new infections in the window by the 95% band; MAE of the regenerated daily new infections in the window; and MAE of the regenerated number currently infectious (not observed by any method). Dashed line: 95%.

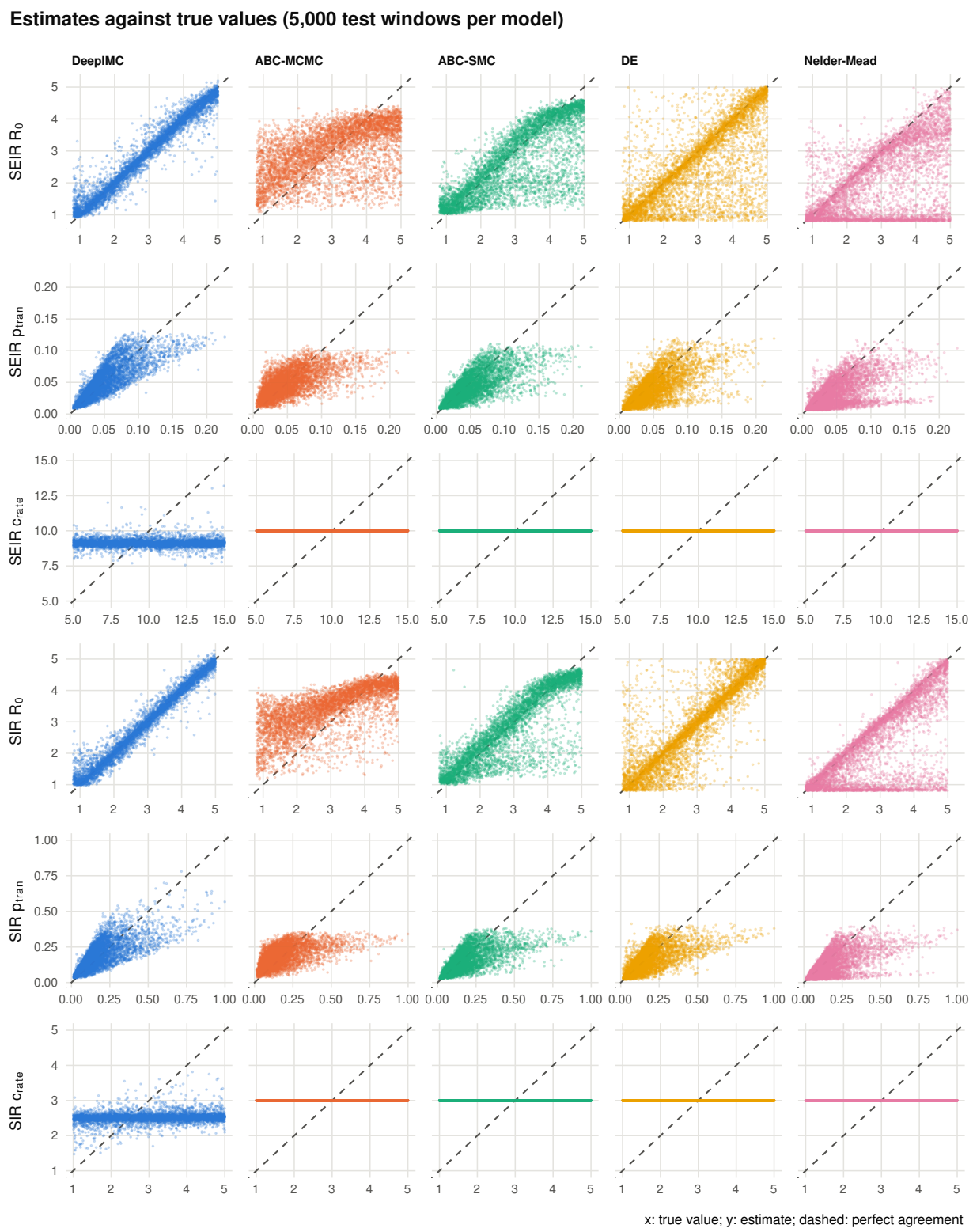


Figure 4: Estimated against true values of R_0, p_{tran} and c_{rate} for every simulated test window and method; the dashed line is perfect agreement. DeepIMC's flat cloud for c_{rate} shows that the curve does not reveal it; the comparison methods fix it at the middle of its range.

Figures 5–7).

New infections in the window. All methods except Nelder–Mead reproduced the curve they had seen. The MAE of the regenerated median curve was 4.8 daily new infections for both DeepIMC and differential evolution in the SEIR windows (6.1 and 6.2 in the SIR windows), against 7.8 to 18.8 for the ABC methods (14.1 to 28.2) and 8.9 for Nelder–Mead (10.6). Differential evolution, which fits each window directly, came as close to the data as DeepIMC, and slightly closer in middle and late windows (Figure 5c, ratios just above 1). DeepIMC's bands covered 98–99% of the days, those of differential evolution 96%, those of the ABC methods 99–100% and those of Nelder–Mead 79–82%.

The number currently infectious. This curve, which no method observes, depends on the estimated starting state as well as on R_0, and it is where the methods differed most. DeepIMC's bands covered the true number of infectious people on 98–99% of the days, and its median curve was off by 17 people on average in the SEIR windows and 30 in the SIR windows—about half the error of the best comparison method, differential evolution (35 and 46). Differential evolution and Nelder–Mead, which return a single estimate, covered the true number on only 81% and 56–67% of the days, and below 80% in a third of the windows for differential evolution and in half or more for Nelder–Mead (Figure 7). The ABC methods covered it on 99–100% of the days, with errors three to eight times larger than DeepIMC's.

The next 14 days. Two weeks beyond the window, the curves regenerated from DeepIMC, differential evolution and ABC-SMC were about equally accurate (MAE 1.7, 1.8 and 2.1 new infections per day for SEIR; 0.7, 0.7 and 0.8 for SIR), and all methods except Nelder–Mead covered 97–100% of the days. Many test windows end when the epidemic is already declining, so the counts over these days are small. Figure 6 shows three typical SEIR windows and the distribution of the errors over all windows.

4.4 Real data

On real data the question is how well each method's regenerated curves match the observed counts, in the window and over the following two weeks. Table 8 and Figure 8 give the coverage and the mean absolute error for each dataset.

Fit to the window. DeepIMC's bands covered 77–100% of the observed days, against 89–100% for the two ABC methods, 66–97% for differential evolution and 42–90% for Nelder–Mead. Coverage alone, however, rewards wide bands: ABC-MCMC reached 90–100% with

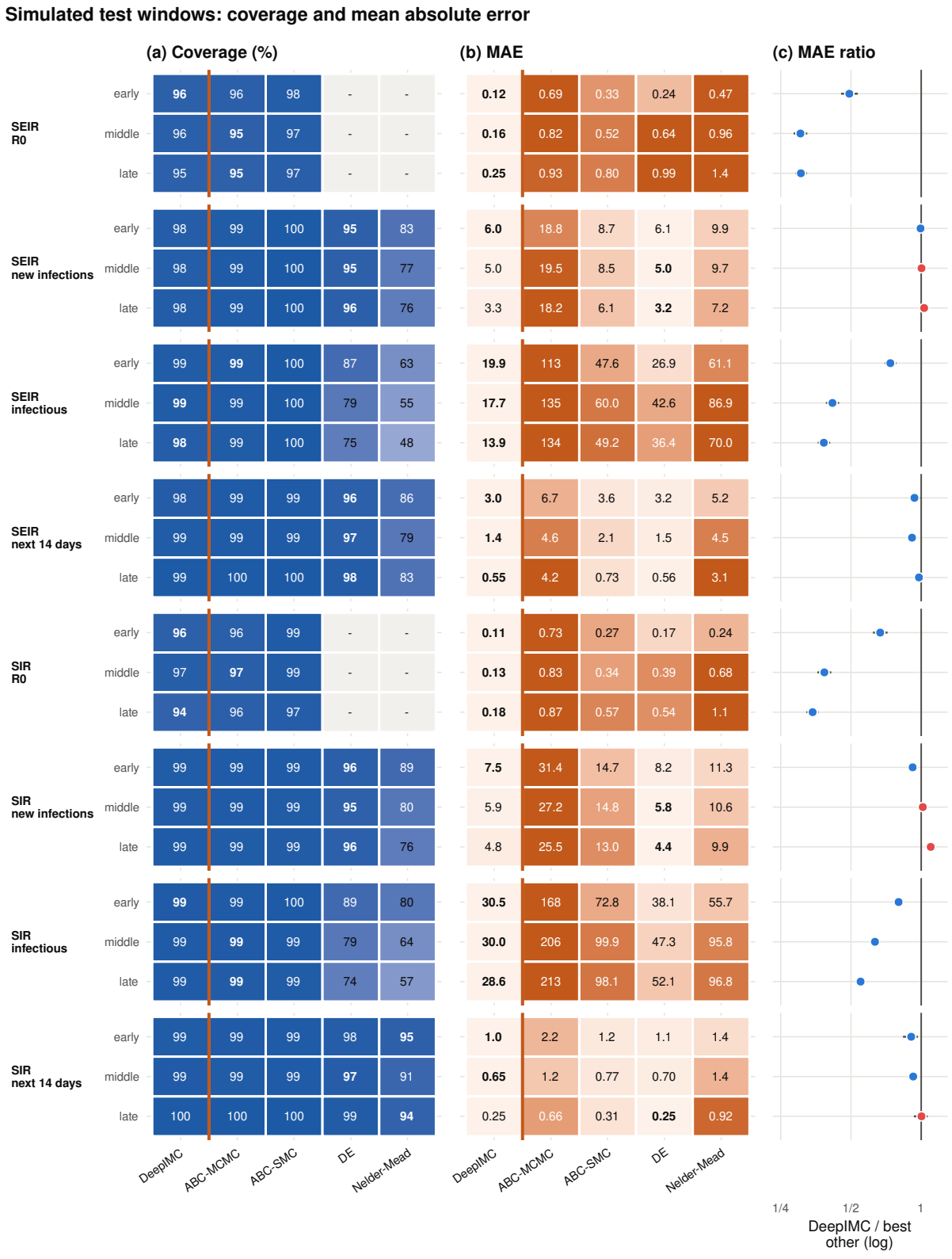


Figure 5: Simulated test windows. (a) Mean coverage by method, for early, middle and late windows: of the true R_0 by the 95% range, and of the true curves by the 95% band (bold: closest to 95%; "-": the method gives no range). (b) Mean absolute error of R_0 and of the regenerated curves (bold: lowest). (c) MAE of DeepIMC divided by that of the best comparison method for the same windows (log scale; below 1: DeepIMC better), with 95% paired bootstrap intervals.

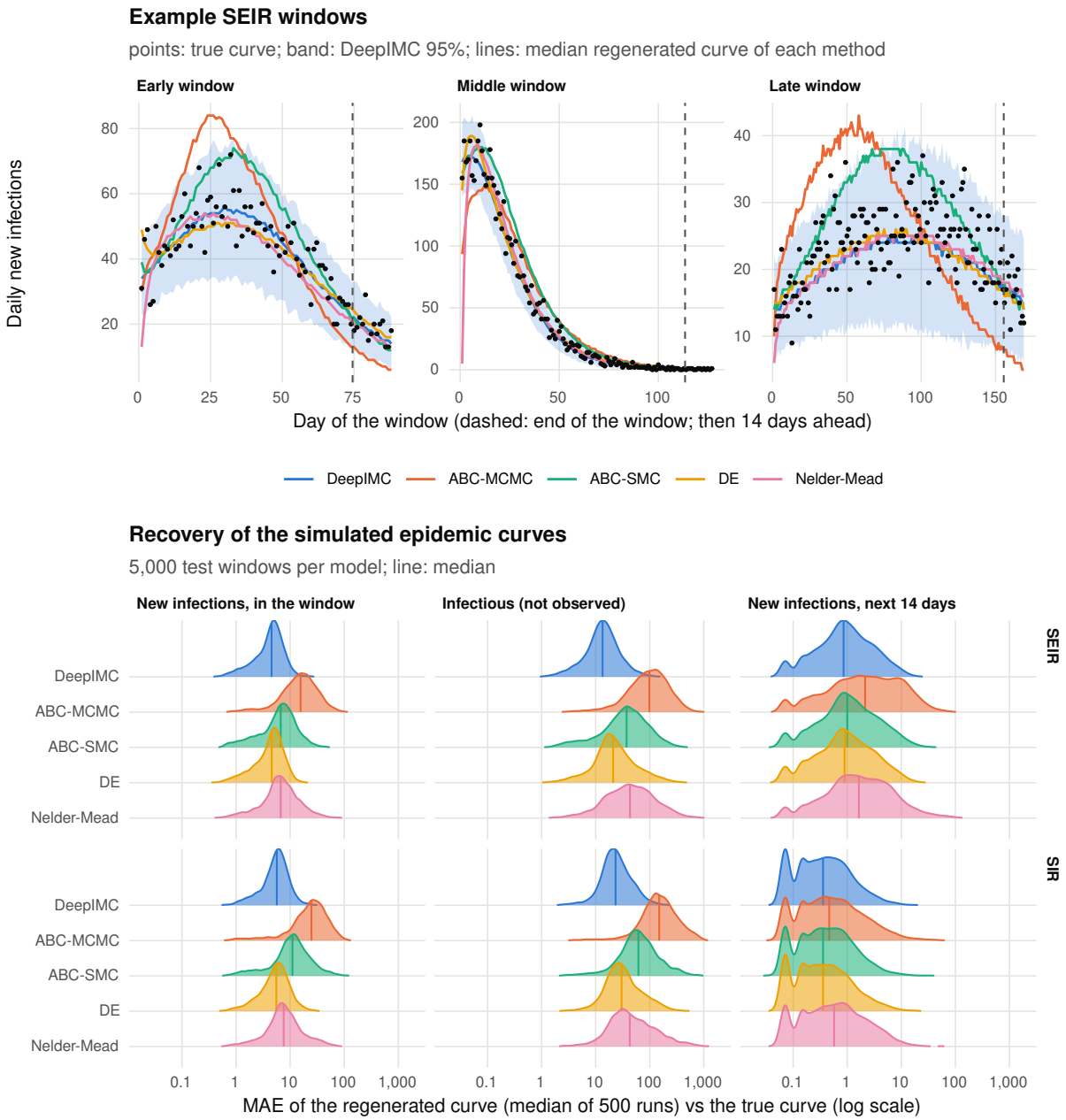


Figure 6: Recovery of the simulated epidemic curves. Top: three SEIR test windows (early, middle and late) with the true curve (points), DeepIMC's 95% band and every method's median regenerated curve; the dashed line marks the end of the window. Bottom: distribution over all test windows of the MAE of the median regenerated curve (log scale; vertical line: median).

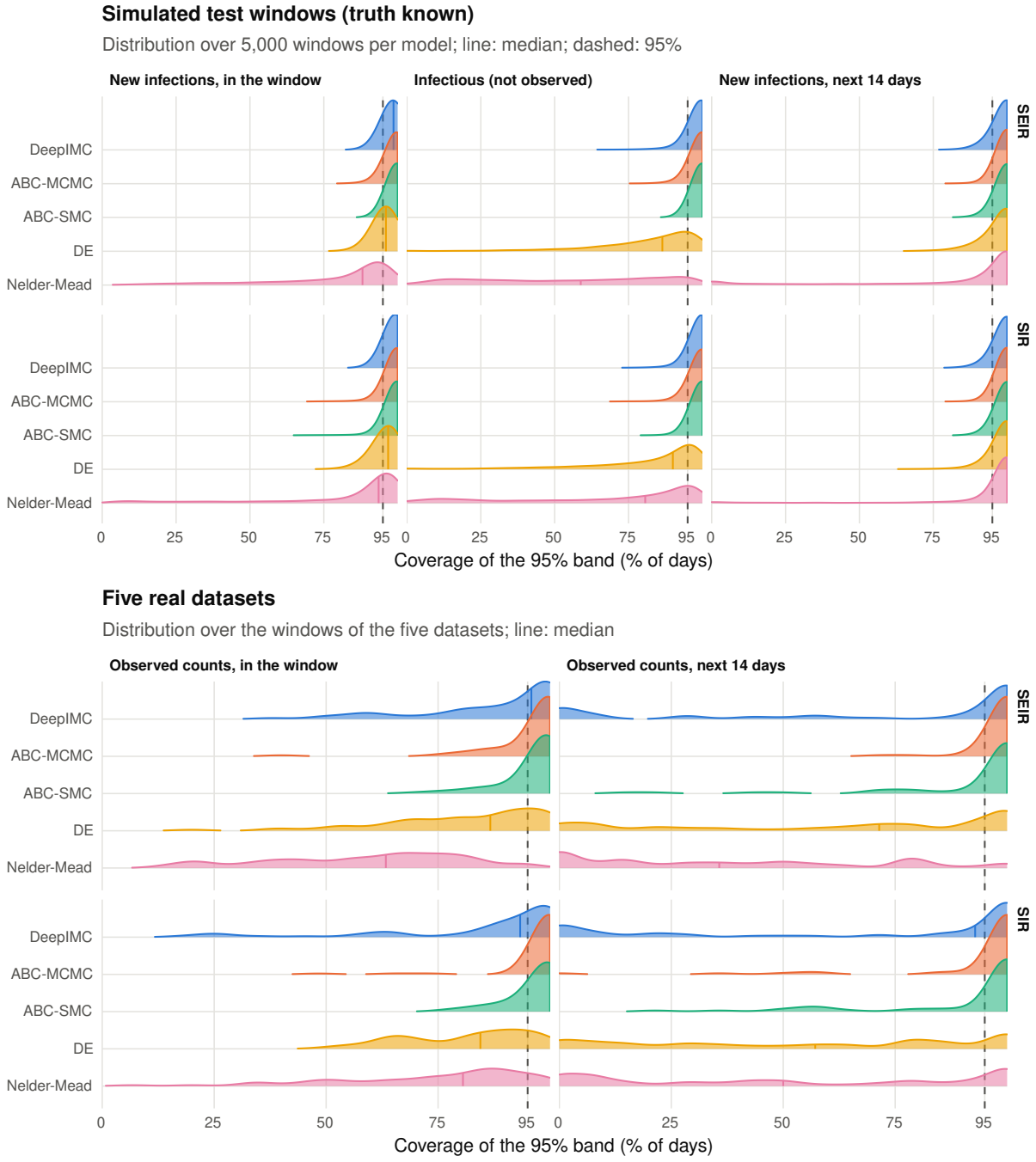


Figure 7: Distribution of the coverage of the 95% band over the windows, by method: simulated test windows (top) and the five real datasets (bottom). Vertical line in each ridge: median; dashed: 95%.

Table 7: Recovery of the epidemic curves on the 10,000 simulated test windows: coverage (%) and MAE of the true daily new infections in the window, of the true number currently infectious in the window (not observed by any method), and of the daily new infections in the 14 days after the window, from 500 model runs per window and method.

		New infections, window		Infectious, window		New infections, next 14 days	
Model	Method	Coverage (%)	MAE	Coverage (%)	MAE	Coverage (%)	MAE
SEIR	DeepIMC	98.1	4.8	**98.4**	**17.2**	98.8	**1.7**
	ABC-MCMC	99.1	18.8	99.0	126.9	99.4	5.2
	ABC-SMC	99.7	7.8	99.8	52.2	99.5	2.1
	DE	**95.5**	**4.8**	80.6	35.2	**96.9**	1.8
	Nelder–Mead	78.6	8.9	55.5	72.4	82.8	4.3
SIR	DeepIMC	98.9	**6.1**	99.0	**29.7**	99.1	**0.7**
	ABC-MCMC	99.0	28.2	**98.9**	194.7	99.2	1.4
	ABC-SMC	99.0	14.1	99.2	89.2	99.5	0.8
	DE	**95.7**	6.2	81.1	45.6	98.0	0.7
	Nelder–Mead	82.0	10.6	67.3	81.3	**93.8**	1.3

bands so wide that its median curve was the furthest from the data on most datasets. By the mean absolute error of the median curve, differential evolution, which fits each window directly, came closest to the data on most datasets, and DeepIMC was close to it on the influenza series, Baltimore 1918 and Ebola (Figure 8). DeepIMC was furthest behind on the Hong Kong SARS outbreak and on the Utah COVID-19 admissions.

By window length and position. Figure 9 shows the same comparison by window length (30, 60 and 90 days) and position. For most methods the coverage of the window fell as the window grew longer, because a longer window is more likely to contain changes that a model with constant parameters cannot follow; DeepIMC's coverage fell most, from 93–100% for 30-day windows to 60–81% for 90-day windows, while its MAE stayed within a factor of two of the best method. The ABC methods kept their coverage with wide bands, at the cost of the largest errors (ABC-MCMC up to 13 times the best).

The next 14 days. Predicting the two weeks after a window was harder for every method. Thanks to their wide bands the ABC methods covered most of these days, while DeepIMC covered 73–100% of them for the influenza series but only 42–52% for the COVID-19 admissions and 32–64% for the SARS and Ebola outbreaks under the SEIR model (Table 9). By the MAE, differential evolution or ABC-SMC came closest on most datasets; DeepIMC was close to them on the influenza series and furthest behind on the SARS outbreak under the SEIR model. Figure 10 shows why: around the peak of the SARS outbreak, DeepIMC continued the growth it saw in the window, while the outbreak, brought under control, de-

clined. A model with constant parameters within a window cannot anticipate such a change; the comparison methods share this limitation, and their wider bands absorb more of it.

Estimated R_0. The median estimates of R_0 for each dataset were between 1.0 and 2.1 for DeepIMC, 1.2 and 2.1 for ABC-SMC and 1.1 and 3.6 for differential evolution. Those of ABC-MCMC (1.9 to 2.9) stayed close to the middle of its prior range, a sign, as on simulated data, that its posterior was close to the prior.

The contrast series. The Utah COVID-19 cases by test report date are shown at the bottom of Table 8. Scaled to cases per 10,000 residents, their daily counts are small (below 40), so their coverage and errors cannot be compared directly with those of the admissions series of the same waves.

Table 8: Real data: coverage (%) of the observed daily counts in the window by the 95% band, and MAE of the median regenerated curve, mean over the windows of each dataset (number of windows per model in brackets). Bold: coverage closest to 95% and lowest MAE. †Contrast series: cases by test report date, per 10,000 residents.

		Coverage of the observed counts (%)					MAE of the median curve				
Dataset	Model	DeepIMC	ABC-MCMC	ABC-SMC	DE	Nelder–Mead	DeepIMC	ABC-MCMC	ABC-SMC	DE	Nelder–Mead
Utah COVID-19 admissions (27)	SEIR	85	100	**98**	82	71	12.7	24.9	13.4	**9.5**	12.1
	SIR	77	100	**96**	81	79	17.2	24.6	14.6	**10.8**	11.4
Utah influenza admissions (18)	SEIR	100	**95**	100	97	46	2.2	26.5	3.3	**2.2**	3.0
	SIR	**96**	98	100	88	90	2.5	15.8	4.2	**2.2**	2.2
Baltimore influenza 1918 (9)	SEIR	85	92	**95**	66	42	36.2	76.9	34.6	**32.6**	41.4
	SIR	88	90	**95**	81	44	32.3	74.9	48.9	**30.3**	50.4
Hong Kong SARS 2003 (9)	SEIR	78	**91**	89	67	59	24.2	30.1	10.8	**8.8**	10.1
	SIR	81	100	**93**	73	60	10.9	18.7	10.6	**8.4**	10.4
Sierra Leone Ebola 2014-15 (9)	SEIR	84	99	**96**	74	65	7.8	18.4	9.2	**6.9**	7.9
	SIR	88	100	**98**	77	73	8.4	15.2	8.2	**7.2**	8.2
Utah COVID-19 cases† (27)	SEIR	**96**	100	100	91	81	4.8	15.5	3.2	**2.8**	3.4
	SIR	**97**	100	100	90	88	3.6	8.9	**3.2**	3.3	3.5

Table 9: Real data: coverage (%) of the observed counts in the 14 days after the window (where the data continue) and MAE of the median regenerated curve over those days. Bold: coverage closest to 95% and lowest MAE.

		Coverage, next 14 days (%)					MAE, next 14 days				
Dataset	Model	DeepIMC	ABC-MCMC	ABC-SMC	DE	Nelder–Mead	DeepIMC	ABC-MCMC	ABC-SMC	DE	Nelder–Mead
Utah COVID-19 admissions	SEIR	52	98	**93**	47	32	22.6	30.2	**17.7**	18.5	24.2
	SIR	42	**92**	88	34	29	27.1	35.6	**21.6**	23.6	25.6
Utah influenza admissions	SEIR	100	100	100	**95**	48	2.1	3.1	2.4	3.6	**1.7**
	SIR	93	**94**	100	82	87	1.7	1.6	1.5	**1.5**	2.1
Baltimore influenza 1918	SEIR	87	**98**	99	75	58	32.6	44.0	31.0	**29.3**	48.1
	SIR	73	**95**	96	81	73	42.6	41.2	34.0	**24.9**	36.2
Hong Kong SARS 2003	SEIR	32	**100**	77	25	17	93.4	**14.9**	28.5	29.3	17.3
	SIR	98	100	**93**	39	30	15.2	**11.2**	21.2	14.1	12.8
Sierra Leone Ebola 2014-15	SEIR	64	**100**	81	58	36	16.1	30.0	18.7	**12.5**	16.3
	SIR	62	**100**	77	57	49	17.0	29.3	18.6	**12.8**	17.7
Utah COVID-19 cases	SEIR	87	100	**98**	80	61	12.8	5.3	**3.5**	5.0	5.3
	SIR	100	100	**99**	79	65	4.5	**3.4**	3.6	6.5	5.7

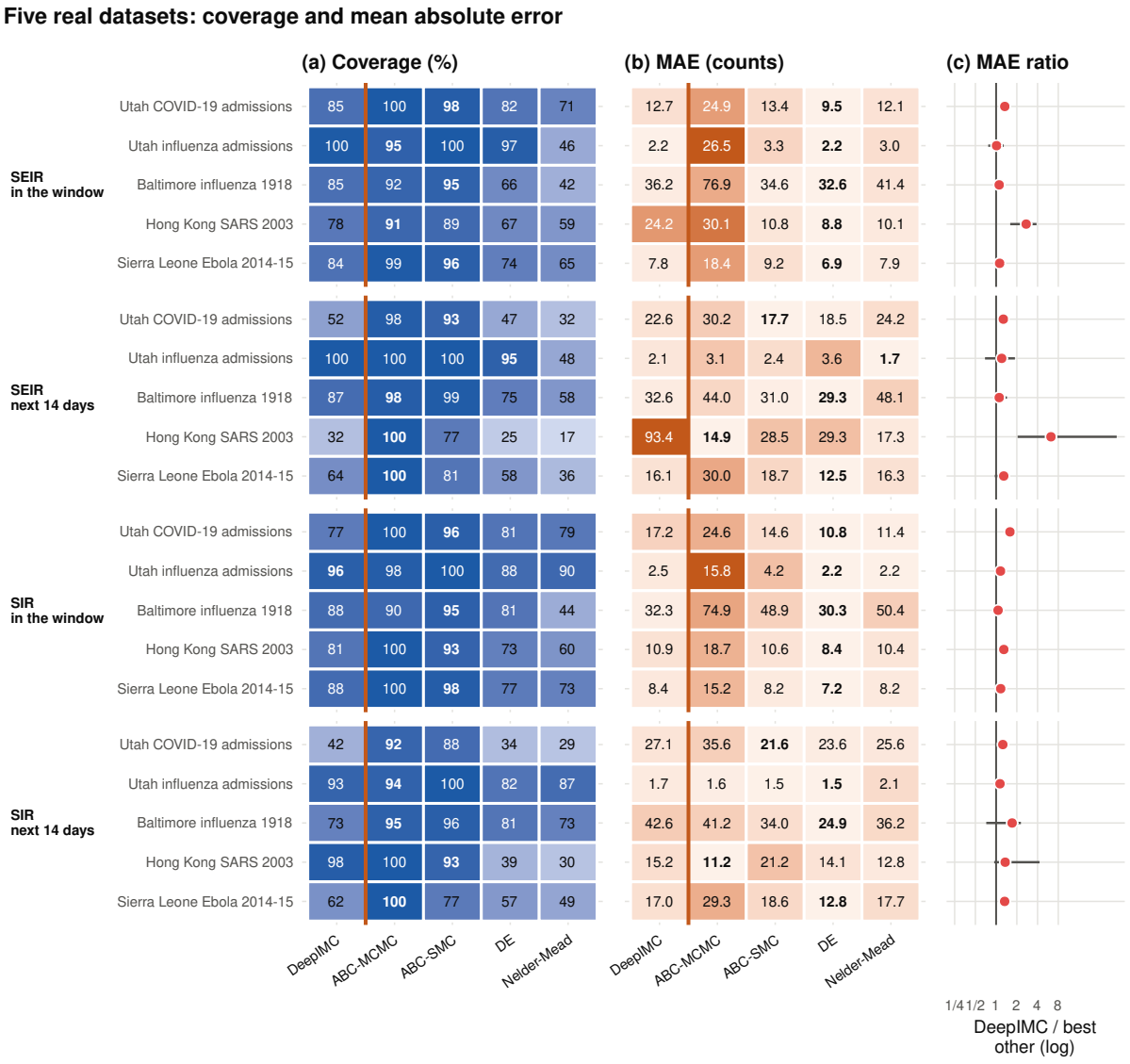


Figure 8: Real data. (a) Mean coverage of the 95% band of each method, for each dataset and model, in the window and in the 14 days after it (bold: closest to 95%). (b) Mean absolute error of the median regenerated curve, in daily counts (bold: lowest). (c) MAE of DeepIMC divided by that of the best comparison method for the same windows (log scale; below 1: DeepIMC better), with 95% paired bootstrap intervals.

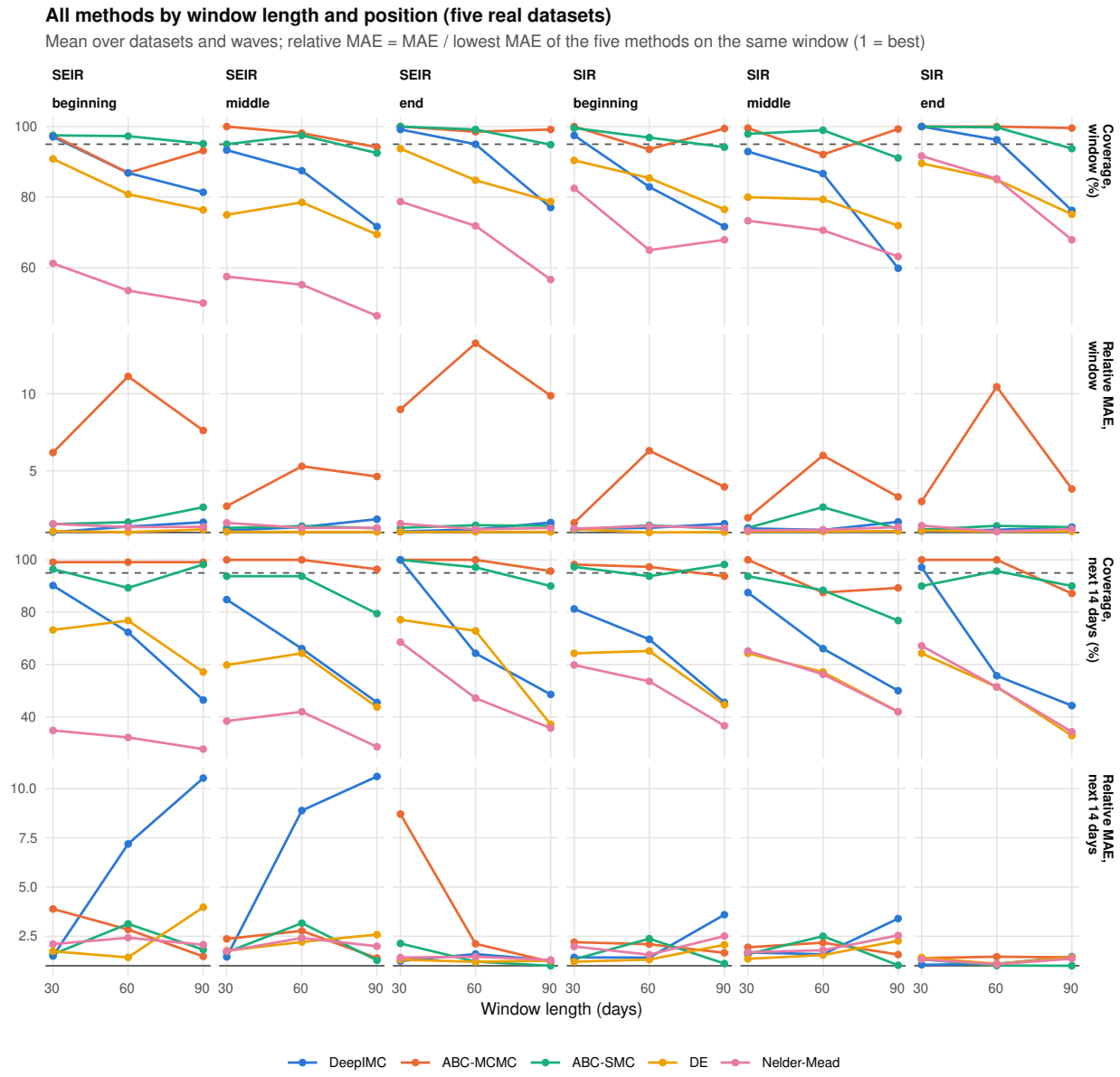


Figure 9: All methods on the five real datasets, by window length and position: coverage of the observed counts by the 95% band and relative MAE of the median regenerated curve (the MAE divided by the lowest MAE of the five methods on the same window, so that datasets of different size can be averaged; 1 is best), in the window and in the 14 days after it.

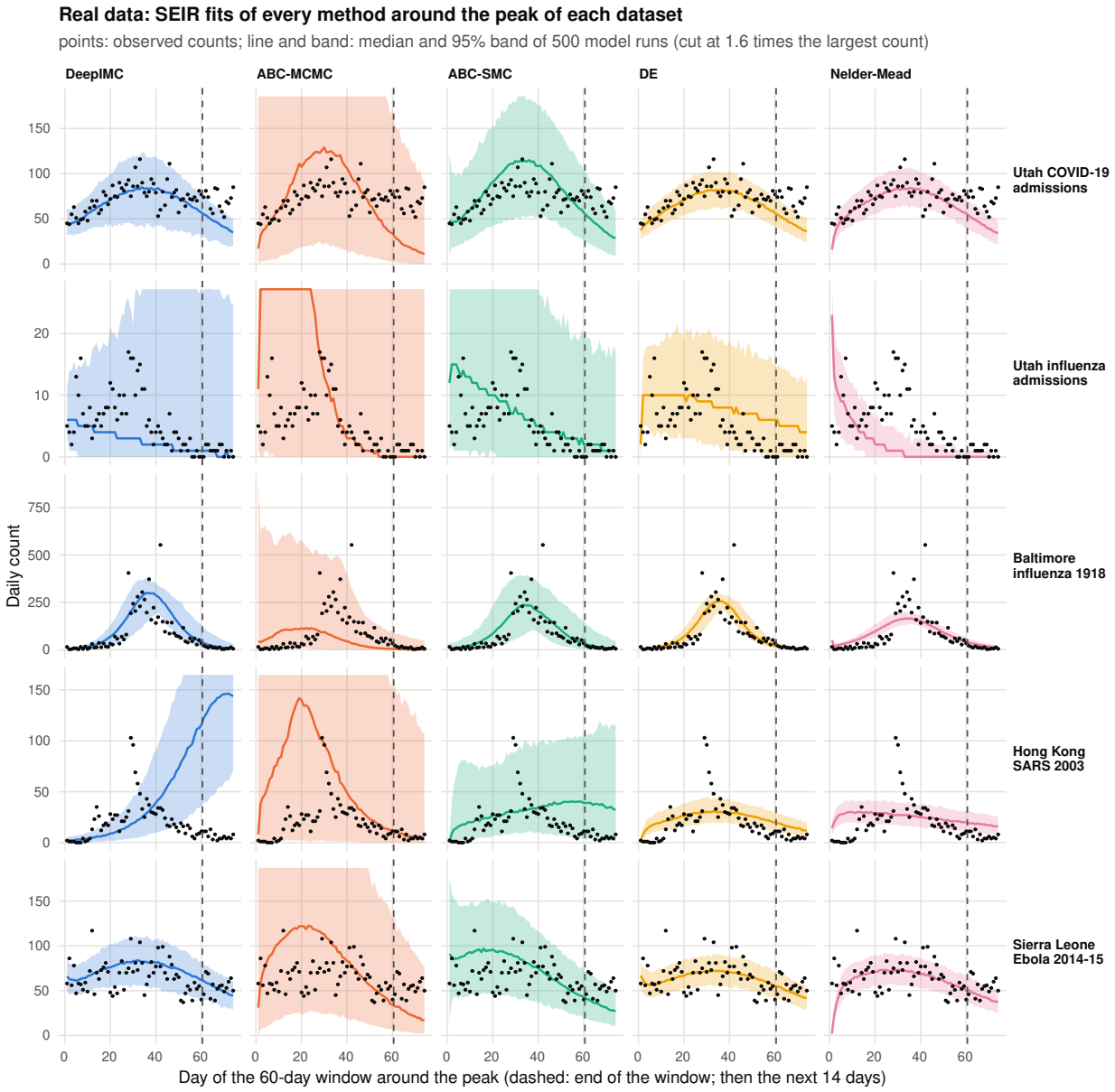


Figure 10: Real data: SEIR fits of every method to the 60-day window in the middle of each dataset, with the next 14 days (after the dashed line). Points: observed counts; lines and bands: median and 95% band of 500 model runs from each method's estimates (cut at 1.6 times the largest observed count). The SIR fits are in Figure 11.

4.5 Computing time

The last question is what each method costs. Table 10 separates the cost of calibrating a new window from DeepIMC's one-time cost of training.

Calibrating a window. On one CPU core, DeepIMC calibrated a window in 0.14 seconds (SEIR) and 0.10 seconds (SIR), and slightly less per window when several windows were processed together. Its first call in an R session took 10 and 6.5 seconds, almost all of it to start Python and load the five networks; this is paid once per session. The comparison methods took between 3.6 seconds (Nelder–Mead, SIR) and 536 seconds (differential evolution, SEIR) per window. For SEIR windows, DeepIMC was about 1,400 times faster than ABC-MCMC, 2,800 times faster than ABC-SMC and 3,800 times faster than differential evolution; it was 70 times faster than Nelder–Mead, the cheapest of the comparison methods.

From a fit to a band. Turning a fit into a 95% band with 500 runs of the agent-based model took 32 (SEIR) and 17 (SIR) seconds per window, far longer than DeepIMC's fit itself. The renewal equation gave a band in 0.18 and 0.11 seconds.

The one-time cost. Simulating the 1,000,000 training epidemics took 30 (SEIR) and 14 (SIR) CPU-hours, and computing DeepIMC's estimates for the 100,000 calibration windows of the R_0 range 1.5 and 0.2 hours on one core. Training is the largest part: one SEIR network would need about 85 hours on 16 CPU cores (30 hours for SIR), which is why we trained on GPUs, where one network took 3.7 to 20 hours depending on the GPU. The simulation time alone equals the cost of calibrating about 200 SEIR windows with differential evolution or 270 with ABC-SMC; with training on CPUs, the one-time cost equals tens of thousands of windows. It is paid once, by the developer: the trained networks are shipped with the package, and its users pay only the cost of a calibration.

5 Discussion

We introduced DeepIMC, a neural network that maps a window of daily new infections of any length, from any part of an epidemic, to the basic reproduction number and the current state of an agent-based SEIR or SIR epidemic. On 10,000 simulated test windows, where the truth is known, it estimated R_0 about three times more accurately than four standard calibration methods used with their default settings. Its ranges for R_0 had close to nominal coverage while being far narrower than those of ABC; it recovered the state of the epidemic, and the number of infectious people that no method observes, better than every other method; and it

Table 10: Computing time on CPUs (one core per method, the same computer for all methods), and the one-time cost of DeepIMC.

	SEIR	SIR
Calibrating one window (seconds, one CPU core)		
DeepIMC, first call (starts Python, loads networks)	10.1	6.5
DeepIMC, one window	0.142	0.099
DeepIMC, 10 windows in one pass, per window	0.120	0.065
ABC-MCMC	199.9	93.2
ABC-SMC	397.4	131.8
DE	536.5	203.1
Nelder–Mead	10.0	3.6
From a fit to a 95% band (seconds per window)		
500 runs of the agent-based model	32.3	16.6
Renewal equation	0.179	0.109
One-time cost of DeepIMC (CPU)		
Simulating 1,000,000 training epidemics (CPU-hours)	29.6	14.1
Estimates for 100,000 calibration windows (hours, 1 core)	1.5	0.2
Training one network, 50 epochs (hours, 16 cores, estimated)	84.9	29.6
For reference: training as done, on GPUs		
Training one network, 50 epochs (hours, one GPU)	3.7–20.4	2.6–9.9

calibrated a window in a fraction of a second on one CPU core. On real data the comparison was closer, and depended on how closely each dataset matched the model.

5.1 What the curve can and cannot tell

The transmission probability and the contact rate cannot be separated from a curve of daily infections, by DeepIMC or by any other method: their errors equal those of a guess that knows R_0 exactly, because only their product shapes the epidemic. This does not limit the use of a calibrated model for prediction, which depends on R_0 and the current state; but estimating p_{tran} and c_{rate} separately requires additional information, such as contact rates from a contact survey.

5.2 Why DeepIMC is more accurate

At the beginning of an epidemic, the growth of the curve reveals R_0 almost directly, and differential evolution came closest to DeepIMC. In the middle and at the end of an epidemic, a method must also find how many people are already infected or immune, and many combinations of R_0 and starting state produce similar curves. The comparison methods search

this space by trial and error, over thousands of stochastic simulations, for every window; DeepIMC has learned from a million simulated epidemics which combinations are likely, and this is where its advantage was largest.

The comparison methods failed in different ways. Differential evolution matched the observed curve as closely as DeepIMC, because it fits each window directly, but its single estimate could not express how uncertain the starting state is, and its estimate of the unobserved number of infectious people was twice as far from the truth. ABC-MCMC, with its tolerance set from 200 simulations from the prior, stayed close to the prior: its estimates were pulled towards the middle of the range and its ranges covered most of it. ABC-SMC was the most accurate comparison method for SEIR windows but still underestimated many large values of R_0. Nelder–Mead was the weakest method on the curves: with a fixed-seed agent-based model the fit is flat and irregular, so its stopping rule is met after about 100 runs, often far from the best values.

5.3 Ease of use

DeepIMC needs only the daily counts and the known values. Each comparison method required choices before it could be used: ranges for R_0 and for the starting state, how to parameterize them, the number of model runs, tolerances and, for ABC, the summary statistics and their scaling. Even with default settings, ABC-SMC failed on most SEIR windows until its limits were raised. These choices are made once, by the developer, when DeepIMC is trained; users of the trained models do not face them.

5.4 Real data and the choice of data

DeepIMC was trained on noise-free epidemics, in which every new infection is counted, in a closed population with constant parameters. Real data depart from these assumptions in three ways: only a fraction of infections is recorded, the recorded counts depend on testing and reporting, and the course of an epidemic changes with behaviour and interventions. The choice of data therefore matters. Counts by date of symptom onset and hospital admissions by date of admission follow new infections more closely than cases by date of report, which follow the reporting week and the availability of tests. On the datasets closest to the model's assumptions—influenza waves recorded by admission or onset, and the Ebola outbreak—DeepIMC fitted the data almost as closely as the best comparison method, in a fraction of a second. On outbreaks that were brought under control, such as SARS in Hong Kong, no method with constant parameters can follow the decline, and DeepIMC, whose bands are narrower, was penalized most.

A version of DeepIMC trained on simulated *reported* cases, with a reported share of infections, weekday effects and extra noise, is a natural way to close this gap. In earlier experiments with 61-day windows, such a version raised the coverage of 24 Utah COVID-19 case-report windows from 46% to 69%: an improvement, but not enough to match the methods that fit each curve directly on data that also change over time in ways a model with constant parameters cannot follow.

5.5 Computational efficiency

Because DeepIMC calibrates a window in a fraction of a second on an ordinary CPU, it can be used where the standard methods are impractical: for many locations at once, every day of an ongoing outbreak as new data arrive, or inside a larger analysis that calibrates thousands of windows, such as the evaluation in this paper. The cost is moved to training, which in practice needs a GPU and is paid once for each model structure and range of parameters. For a model that will be calibrated only a few times, a standard method is cheaper; for a model that is calibrated repeatedly, as in surveillance or by the many users of a shipped model, the one-time cost is soon recovered. Once the calibration takes a fraction of a second, the slowest step of a prediction is the simulation of the band.

The renewal equation reduces that step to a fraction of a second (Appendix F): started from DeepIMC's estimated state, it gave nearly the same curves of new infections as 500 model runs, about 180 times faster, at the price of bands for the number currently infectious that are too narrow.

5.6 Limitations

The models assume homogeneous mixing, parameters that are constant within a window, and a known population size, recovery rate and latent period. Real population sizes must be mapped to the trained range, and the reported fraction of infections cannot be separated from the population size. The trained ranges of Table 1 bound the epidemics the shipped models can calibrate: the shortest infectious period is 4 days, longer than that of influenza, so influenza is represented with a longer generation interval than observed. Other settings require new training data and retraining, which the software supports. Finally, the R_0 ranges are calibrated empirically on simulated windows rather than derived from a posterior distribution.

5.7 Future work

Training DeepIMC on simulated reported data—with a reported fraction of infections, a weekly reporting pattern and extra noise—would let it read case reports directly. Parameters that change over time, other contact structures such as networks, and models with hospitalization are natural extensions; the same training code applies to any model that can be simulated.

6 Conclusions

Calibrating agent-based models is computationally demanding and requires many choices from the user. DeepIMC reverses the usual approach: it learns, once, the inverse mapping from epidemic curves to the quantities that generated them, using data simulated from the model itself. For agent-based SEIR and SIR models, it estimates R_0 and the current state of an epidemic from a window of any length and position, reproduces the epidemic those estimates imply, and answers in a fraction of a second on an ordinary CPU, with no settings to choose. How well it does on real data depends on how closely the data resemble the simulations it was trained on, which makes the choice of data—onset dates and admissions rather than reported cases—as important as the choice of method.

7 Software availability

The R package `epiworldRcalibrate` provides:

- the trained SEIR and SIR calibrators: `window_calibrator()` and `calibrate()`;
- the four comparison methods, set up as in this paper: `window_baselines()`, with `epidemic_summary()` for the summary statistics of the ABC methods;
- predictions by model simulation or the renewal equation: `predict_cases()`;
- the evaluation scores: `coverage()`, `interval_score()` and `score_band()`;
- the comparison of methods on many windows: `compare_methods()`;
- simulation of training data, tuning and training of new calibrators: `simulate_window_data()`, `tune_window_calibrator()` and `train_window_calibrator()`.

All code needed to reproduce the results is available at: repository link removed for double-blind review.

Acknowledgments

Acknowledgments removed for double-blind review.

References

[1] Takuya Akiba et al. "Optuna: A Next-generation Hyperparameter Optimization Framework". In: *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. 2019, pp. 2623–2631. DOI: 10.1145/3292500.3330701.

[2] Claudio Angione, Eric Silverman, and Elisabeth Yaneske. "Using Machine Learning as a Surrogate Model for Agent-Based Simulations". In: *PLoS ONE* 17.2 (2022), e0263150. DOI: 10.1371/journal.pone.0263150.

[3] Mark A. Beaumont, Wenyang Zhang, and David J. Balding. "Approximate Bayesian Computation in Population Genetics". In: *Genetics* 162.4 (2002), pp. 2025–2035. DOI: 10.1093/genetics/162.4.2025.

[4] Mark A. Beaumont et al. "Adaptive Approximate Bayesian Computation". In: *Biometrika* 96.4 (2009), pp. 983–990. DOI: 10.1093/biomet/asp052.

[5] Anne Cori et al. "A New Framework and Software to Estimate Time-Varying Reproduction Numbers During Epidemics". In: *American Journal of Epidemiology* 178.9 (2013), pp. 1505–1512. DOI: 10.1093/aje/kwt133.

[6] Anne Cori et al. "Temporal Variability and Social Heterogeneity in Disease Transmission: The Case of SARS in Hong Kong". In: *PLoS Computational Biology* 5.8 (2009), e1000471. DOI: 10.1371/journal.pcbi.1000471.

[7] Kyle Cranmer, Johann Brehmer, and Gilles Louppe. "The Frontier of Simulation-Based Inference". In: *Proceedings of the National Academy of Sciences* 117.48 (2020), pp. 30055–30062. DOI: 10.1073/pnas.1912789117.

[8] Christl A. Donnelly, Azra C. Ghani, Gabriel M. Leung, et al. "Epidemiological Determinants of Spread of Causal Agent of Severe Acute Respiratory Syndrome in Hong Kong". In: *The Lancet* 361.9371 (2003), pp. 1761–1766. DOI: 10.1016/S0140-6736(03)13410-1.

[9] Li-Qun Fang, Yang Yang, Jia-Fu Jiang, et al. "Transmission Dynamics of Ebola Virus Disease and Intervention Effectiveness in Sierra Leone". In: *Proceedings of the National Academy of Sciences* 113.16 (2016), pp. 4488–4493. DOI: 10.1073/pnas.1518587113.

[10] Christophe Fraser. "Estimating Individual and Household Reproduction Numbers in an Emerging Epidemic". In: *PLoS ONE* 2.8 (2007), e758. DOI: 10.1371/journal.pone.0000758.

[11] Wade H. Frost and Edgar Sydenstricker. "Influenza in Maryland: Preliminary Statistics of Certain Localities". In: *Public Health Reports* 34.11 (1919), pp. 491–504.

[12] Nicolò Gozzi et al. "Preliminary modeling estimates of the relative transmissibility and immune escape of the Omicron SARS-CoV-2 variant of concern in South Africa". In: *medRxiv* (2022), pp. 2022–01.

[13] Sepp Hochreiter and Jürgen Schmidhuber. "Long Short-Term Memory". In: *Neural Computation* 9.8 (1997), pp. 1735–1780. DOI: 10.1162/neco.1997.9.8.1735.

[14] Hawre Jalal, Thomas A. Trikalinos, and Fernando Alarid-Escudero. "BayCANN: Streamlining Bayesian Calibration With Artificial Neural Network Metamodeling". In: *Frontiers in Physiology* 12 (2021), p. 662314. ISSN: 1664-042X. DOI: 10.3389/fphys.2021.662314.

[15] Diederik P. Kingma and Jimmy Ba. "Adam: A Method for Stochastic Optimization". In: *3rd International Conference on Learning Representations (ICLR)*. 2015. arXiv: 1412.6980.

[16] Francesco Lamperti, Andrea Roventini, and Amir Sani. "Agent-Based Model Calibration Using Machine Learning Surrogates". In: *Journal of Economic Dynamics and Control* 90 (May 2018), pp. 366–389. ISSN: 01651889. DOI: 10.1016/j.jedc.2018.03.011.

[17] Francesco Lamperti, Andrea Roventini, and Amir Sani. *Agent-Based Model Calibration using Machine Learning Surrogates*. 2017. arXiv: 1703.10639 [q-fin.EC]. URL: https://arxiv.org/abs/1703.10639.

[18] Justin Lessler et al. "Incubation Periods of Acute Respiratory Viral Infections: A Systematic Review". In: *The Lancet Infectious Diseases* 9.5 (2009), pp. 291–300. DOI: 10.1016/S1473-3099(09)70069-6.

[19] Qun Li, Xuhua Guan, Peng Wu, et al. "Early Transmission Dynamics in Wuhan, China, of Novel Coronavirus–Infected Pneumonia". In: *New England Journal of Medicine* 382.13 (2020), pp. 1199–1207. DOI: 10.1056/NEJMoa2001316.

[20] Marc Lipsitch, Ted Cohen, Ben Cooper, et al. "Transmission Dynamics and Control of Severe Acute Respiratory Syndrome". In: *Science* 300.5627 (2003), pp. 1966–1970. DOI: 10.1126/science.1086616.

[21] Ilya Loshchilov and Frank Hutter. "SGDR: Stochastic Gradient Descent with Warm Restarts". In: *5th International Conference on Learning Representations (ICLR)*. 2017. arXiv: 1608.03983.

[22] Paul Marjoram et al. "Markov Chain Monte Carlo without Likelihoods". In: *Proceedings of the National Academy of Sciences of the United States of America* 100.26 (Dec. 2003), pp. 15324–15328. ISSN: 00278424. DOI: 10.1073/pnas.0306899100. pmid: 14663152.

[23] Derek Meyer and George G Vega Yon. "epiworldR: Fast Agent-Based Epi Models". In: *Journal of Open Source Software* 8.90 (Oct. 3, 2023), p. 5781. ISSN: 2475-9066. DOI: 10.21105/joss.05781. URL: https://joss.theoj.org/papers/10.21105/joss.05781.

[24] Katharine M. Mullen et al. "DEoptim: An R Package for Global Optimization by Differential Evolution". In: *Journal of Statistical Software* 40.6 (2011), pp. 1–26. DOI: 10.18637/jss.v040.i06.

[25] John A. Nelder and Roger Mead. "A Simplex Method for Function Minimization". In: *The Computer Journal* 7.4 (1965), pp. 308–313. DOI: 10.1093/comjnl/7.4.308.

[26] Donovan Platt. "A comparison of economic agent-based model calibration methods". In: *Journal of Economic Dynamics and Control* 113 (2020), p. 103859.

[27] Dennis Prangle. "Adapting the ABC Distance Function". In: *Bayesian Analysis* 12.1 (2017), pp. 289–309. DOI: 10.1214/16-BA1002.

[28] Stefan T. Radev et al. "BayesFlow: Learning Complex Stochastic Models With Invertible Neural Networks". In: *IEEE Transactions on Neural Networks and Learning Systems* 33.4 (2022), pp. 1452–1466. DOI: 10.1109/TNNLS.2020.3042395.

[29] Steven F. Railsback and Volker Grimm. *Agent-Based and Individual-Based Modeling: A Practical Introduction*. 2nd ed. Princeton, NJ: Princeton University Press, 2019. ISBN: 9780691190839.

[30] Rainer Storn and Kenneth Price. "Differential Evolution – A Simple and Efficient Heuristic for Global Optimization over Continuous Spaces". In: *Journal of Global Optimization* 11.4 (1997), pp. 341–359. DOI: 10.1023/A:1008202821328.

[31] Tina Toni et al. "Approximate Bayesian Computation Scheme for Parameter Inference and Model Selection in Dynamical Systems". In: *Journal of the Royal Society Interface* 6.31 (Feb. 2009), pp. 187–202. DOI: 10.1098/rsif.2008.0172.

[32] U.S. Department of Health and Human Services. *COVID-19 Reported Patient Impact and Hospital Capacity by State Timeseries*. 2024. URL: https://healthdata.gov/Hospital/COVID-19-Reported-Patient-Impact-and-Hospital-Capa/g62h-syeh (visited on 10/01/2026).

[33] Utah Department of Health and Human Services. *Utah COVID-19 Data*. 2025. URL: https://coronavirus-dashboard.utah.gov/Utah_COVID19_data.zip (visited on 10/01/2026).

[34] Jacco Wallinga and Marc Lipsitch. "How Generation Intervals Shape the Relationship between Growth Rates and Reproductive Numbers". In: *Proceedings of the Royal Society B: Biological Sciences* 274.1609 (2007), pp. 599–604. DOI: 10.1098/rspb.2006.3754.

[35] WHO Ebola Response Team. "Ebola Virus Disease in West Africa – The First 9 Months of the Epidemic and Forward Projections". In: *New England Journal of Medicine* 371.16 (2014), pp. 1481–1495. DOI: 10.1056/NEJMoa1411100.

[36] Yi Zhang, Zhe Li, Yongchao Zhang, et al. "Validation and calibration of an agent-based model: A surrogate approach". In: *Discrete Dynamics in Nature and Society* 2020 (2020).

A Simulation details

Epidemics were simulated with `ModelSEIRCONN` and `ModelSIRCONN` of **epiworldR** for 365 days with the distributions of Table 1. A new epidemic starts with the drawn share of the population exposed (SEIR) or infectious (SIR) and nobody recovered. The daily count is the number of transitions from exposed to infectious (SEIR) or from susceptible to infectious (SIR); the shares in each state are recorded at the start of each day, so the state at the start of a window is the state at the start of its first day. Separate sets were simulated for training (1,000,000 epidemics per model, 80% for training and 20% for validation), for the calibration windows of the R_0 range (100,000), for the test of DeepIMC by window length (1,000) and for the comparison with the other methods (10,000 in all).

B Network and training details

The hyperparameter search used Optuna's tree-structured Parzen estimator with a median pruner (trials worse than the median of earlier trials after 5 epochs were stopped). Each trial trained one network for 20 epochs on 100,000 training epidemics and was scored by the

Table 11: Settings of the two DeepIMC models. The search space of the hyperparameter search is given in brackets.

Setting	SEIR	SIR
BiLSTM layers [1–3]	3	2
Hidden units per direction [64, 128, 160, 256]	128	64
Dropout between layers [0–0.5]	0.13	0.28
Initial learning rate [10^{-4}–10^{-3}, log scale]	7.16×10^{-4}	7.94×10^{-4}
Batch size [32, 64, 128, 256]	32	32
Learning-rate schedule	cosine, to 1% over 50 epochs	same
Gradient-norm clipping	1	1
Dense layer	64 units, ReLU	same
Output activations	sigmoid (p_{tran}, shares), softplus (R_0)	same
Networks averaged	5	5

validation mean squared error, half for R_0 and half for the state shares. The best settings (Table 11) improved the validation score by 3% (SEIR) and 5% (SIR) over the settings of an earlier model.

Table 12 compares three versions of the networks on the same test windows. *Current* uses the settings of the earlier model, 300,000 training epidemics and 100 epochs with a constant learning rate; *Tuned* uses the settings of Table 11, otherwise as Current; *Final*, the version used in this paper, uses the tuned settings, 1,000,000 epidemics and the cosine learning-rate schedule over 50 epochs. Tuning alone changed the test errors little; more training data and the decreasing learning rate reduced the R_0 error further. As expected, no version improved p_{tran} or c_{rate}, which the curves do not reveal.

Table 12: Mean absolute error of R_0, p_{tran} and c_{rate} of three versions of the networks (5 networks each) on the same test windows: 61-day windows from the beginning of the test epidemics, and all test windows (lengths 30–365 days at the beginning, middle and end).

		61-day windows from the beginning			All test windows		
Model	Version	R_0	p_{tran}	c_{rate}	R_0	p_{tran}	c_{rate}
SEIR	Current	0.121	0.0145	2.58	0.223	0.0141	2.56
	Tuned	0.118	0.0147	2.60	0.225	0.0141	2.57
	Final	0.107	0.0145	2.59	0.218	0.0141	2.57
SIR	Current	0.109	0.0483	1.06	0.192	0.0524	1.05
	Tuned	0.106	0.0490	1.08	0.195	0.0532	1.07
	Final	0.098	0.0486	1.07	0.189	0.0527	1.06

C Comparison methods

All four methods search $\vartheta = (R_0, \log_{10} e_0, \log_{10} i_0, \log_{10} r_0)$ for SEIR and $(R_0, \log_{10} i_0, \log_{10} r_0)$ for SIR, where e_0, i_0 and r_0 are the shares exposed, infectious and recovered on the first day of the window, within $R_0 \in [0.8, 5]$, $\log_{10} e_0, \log_{10} i_0 \in [-4, \log_{10} 0.4]$ and $\log_{10} r_0 \in [-4, \log_{10} 0.95]$. If the shares add up to more than 0.95 they are scaled down to 0.95. The contact rate is fixed at the middle of its training range (10 for SEIR, 3 for SIR) and $p_{tran} = R_0\, p_{recov}/c_{rate}$. For a candidate ϑ, the model is run for the length of the window from the candidate state, and its daily new infectious people $\hat{y}_t(\vartheta)$ are compared with the observed counts y_t: by $\sum_t (y_t - \hat{y}_t(\vartheta))^2$ for DE and Nelder–Mead, and through summary statistics for the ABC methods.

Summary statistics. For a curve $x_1, \ldots, x_T$: $\log(1+\bar{x})$, $\log(1+\max_t x_t)$, $\log(1+\min_t x_t)$, the day of the maximum of the centred 7-day average, $\log(1+\cdot)$ of the mean of the first and of the last 7 days, the first days on which the cumulative count reaches 25%, 50% and 75% of $\sum_t x_t$, and the least-squares slopes of $\log(1 + x_t)$ against t over the first and the last 14 days (function `epidemic_summary()` of the package). With $s(\cdot)$ the vector of statistics and σ_k the median absolute deviation of statistic k over simulations from the prior, the distance is $d = \left(\frac{1}{K} \sum_k [(s_k(\hat{y}) - s_k(y))/\sigma_k]^2\right)^{1/2}$.

ABC-MCMC. 200 simulations from the prior give σ_k and the tolerance ε, the 5% quantile of their distances; Gaussian random-walk proposals with standard deviation 8% of each range, uniform prior on the box, Gaussian kernel $\exp(-d^2/2\varepsilon^2)$, starting at the middle of the box; 3,000 steps, the last 1,500 kept; estimates are posterior medians and the R_0 range the 2.5% and 97.5% posterior quantiles.

ABC-SMC. 200 particles over 6 generations; the first generation, drawn from the prior, gives σ_k; each tolerance is the median distance of the previous generation; perturbation by a multivariate normal kernel with twice the weighted covariance of the particles; at most 30,000 model runs and 20,000 attempts per generation.

DE. `DEoptim` with strategy 2 (DE/local-to-best/1/bin), population 10× the number of unknowns, $F = 0.8$, $CR = 0.5$, 200 generations.

Nelder–Mead. `stats::optim` with its default settings (relative tolerance $\approx 1.5 \times 10^{-8}$, at most 500 function evaluations), started at the middle of the box; candidates outside the box receive an infinite fit.

D ABC on the whole curve or on summary statistics

The ABC methods were also run comparing the whole curve day by day (root mean squared error, with ABC-MCMC's tolerance 0.30× the root mean square of the observed counts) on the 216 windows of an earlier design (9 test epidemics for each model, window length of 30, 61, 120 and 180 days, and position at the beginning, middle and end of the epidemic). Table 13 compares the two versions on the same windows. With default settings, ABC-SMC gave similar results either way, while ABC-MCMC was less accurate on summary statistics, mainly because its fixed tolerance, set from 200 prior simulations, was loose.

Table 13: ABC on the whole curve or on summary statistics, on the same 216 simulated windows: R_0 mean absolute error, starting-state error (percentage points), and coverage and MAE of the true daily new infections in the window.

Model	Method	Compares	R_0 MAE	State error	Coverage (%)	MAE	Seconds
SEIR	ABC-MCMC	whole curve	0.78	9.3	99.2	8.7	73
	ABC-MCMC	summary statistics	0.95	10.0	98.7	15.7	143
	ABC-SMC	whole curve	0.65	7.7	99.1	4.9	306
	ABC-SMC	summary statistics	0.61	6.2	99.6	6.5	292
SIR	ABC-MCMC	whole curve	0.56	7.8	99.3	9.7	56
	ABC-MCMC	summary statistics	1.13	9.7	96.8	19.7	92
	ABC-SMC	whole curve	0.39	7.0	99.2	7.0	169
	ABC-SMC	summary statistics	0.42	5.5	99.6	8.8	171

E Real data: SIR fits

F Supplementary results: predicting with the renewal equation

Running the agent-based model 500 times for every band is the slowest part of a prediction. The renewal equation is a deterministic shortcut that gives the model's average curve.

Renewal A. Started from DeepIMC's estimated state, the renewal equation gave almost the same curves of new infections as 500 runs of the agent-based model (Table 14): in the window, a coverage of 97.5% against 98.1% and an MAE of 5.0 against 4.8 for SEIR (96.2% and 7.8 against 98.9% and 6.1 for SIR), and the same coverage and MAE over the next 14 days. Its bands for the number currently infectious were narrower than those of the model, because they contain only Poisson noise around the average and none of the randomness of

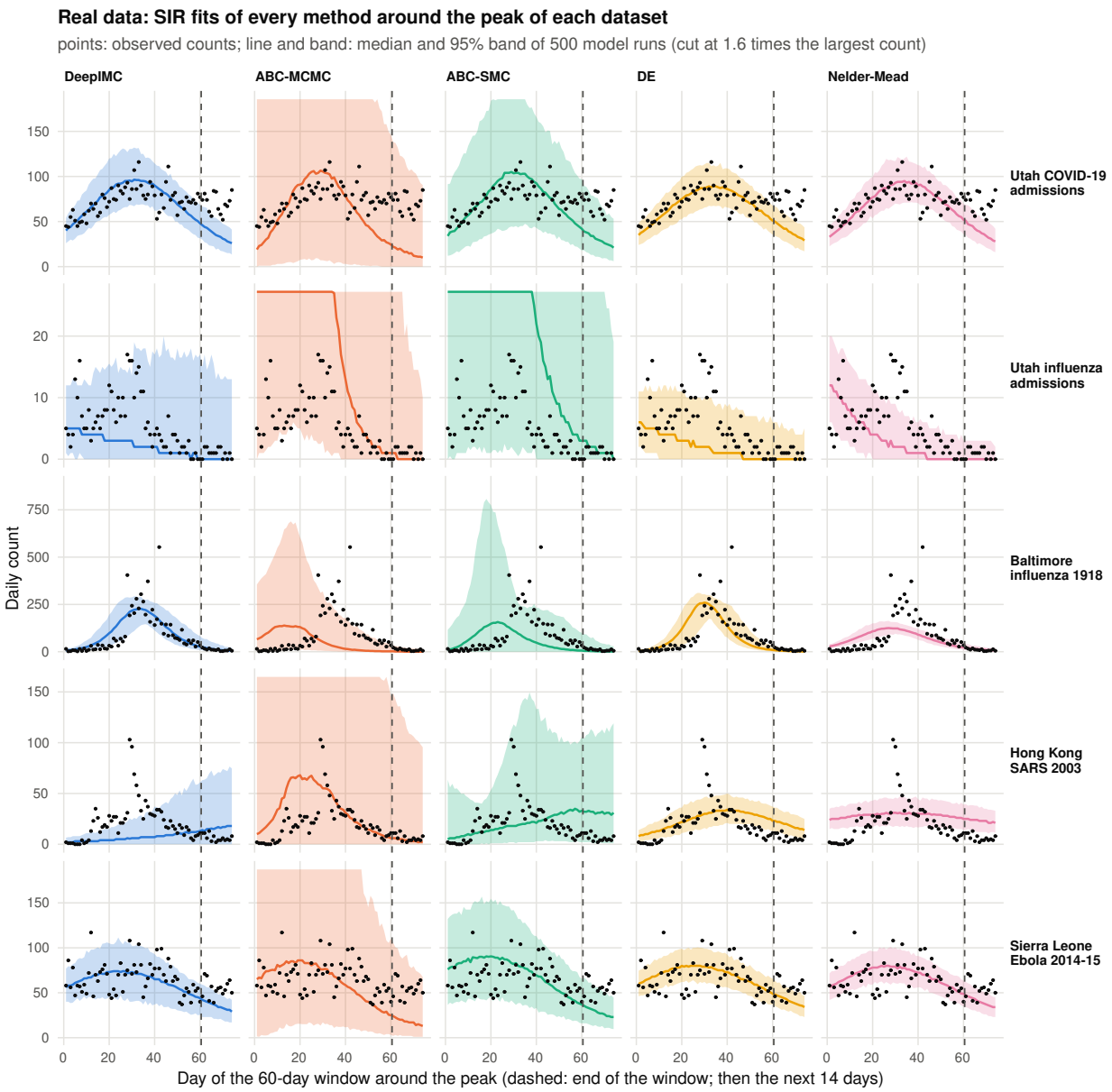


Figure 11: Real data: SIR fits of every method to the 60-day window in the middle of each dataset, as in Figure 10.

the epidemic itself, and covered 86–89% of the days instead of 98–99%. On one CPU core, it took 0.18 seconds per SEIR window instead of 32 seconds for 500 model runs (Table 10).

Renewal B. Fitting the starting state with the renewal equation did not help. The fitted state was less accurate than DeepIMC's (2.0 against 1.4 percentage points for SEIR, 2.7 against 2.2 for SIR). The fitted curve was slightly closer to the observed new infections (MAE 4.4 against 4.8 for SEIR), but from the curve alone the renewal fit cannot separate exposed from infectious people, and its bands covered the true number of infectious people on only 39–47% of the days. Renewal B was also much slower than renewal A, because it fits the state at 30 values of R_0.

Table 14: From DeepIMC's estimates to curves on the simulated test windows: 500 runs of the agent-based model, or the renewal equation started from DeepIMC's state (A) or from a state fitted by the renewal equation (B). Starting-state error (percentage points); coverage (%) of the true curves and MAE of the median curve; seconds per window during the evaluation, on shared computers (Table 10 compares the times on one dedicated core).

		State	New infections		Currently infectious		Next 14 days		Seconds
Model	Curves from	error	Cov.	MAE	Cov.	MAE	Cov.	MAE	per window
SEIR	model, 500 runs	1.4	98.1	4.8	98.4	17.2	98.8	1.7	16.7
	renewal A	1.4	97.5	5.0	88.8	18.6	98.3	1.7	0.90
	renewal B	2.0	96.5	4.4	38.9	21.3	97.4	1.6	68.4
SIR	model, 500 runs	2.2	98.9	6.1	99.0	29.7	99.1	0.7	8.1
	renewal A	2.2	96.2	7.8	85.5	36.7	98.3	0.7	0.49
	renewal B	2.7	95.3	5.3	47.2	30.8	98.0	0.6	6.02

G Renewal equation

With $\beta = R_0\, p_{recov}$ (the daily transmission rate $p_{tran}\, c_{rate}$), $\sigma = 1/\text{incub}$ and $\gamma = p_{recov}$, starting from S_0, E_0, I_0 (the estimated numbers on the first day of the window), we compute for $t = 1, 2, \ldots$

$$\text{new infections}_t = S_{t-1}\left(1 - e^{-\beta I_{t-1}/N}\right),$$
$$C_t = \sigma E_{t-1} \text{ (SEIR)}, \qquad C_t = \text{new infections}_t \text{ (SIR)},$$
$$E_t = E_{t-1} + \text{new infections}_t - C_t \text{ (SEIR)},$$
$$I_t = I_{t-1} + C_t - \gamma I_{t-1}, \qquad S_t = S_{t-1} - \text{new infections}_t.$$

Because the latent and infectious periods are geometric, this recursion equals the renewal equation (2) with generation interval $g_k = \sum_{u+s=k} \sigma(1-\sigma)^{u-1}\,\gamma(1-\gamma)^{s-1}$ for SEIR and $g_k = \gamma(1-\gamma)^{k-1}$ for SIR, with the people already exposed or infectious at the start of the window as its initial condition.